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Paleomagnetism and Neogene Tectonics of the Northern Channel Islands, California

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Geologic observations and previous paleomagnetic studies have suggested that the western Transverse Ranges arrived at their anomalous east-west orientation by clockwise tectonic rotation. A paleomagnetic study of the northern Channel Islands was undertaken, in order to test the extent of rotated areas and to develop constraints on tectonic models concerning the formation of the southern California Borderland. The islands of Anacapa, Santa Cruz, Santa Rosa, and San Miguel form an east-west trending chain and are considered part of the Transverse Range physiographic province. Oligocene through Miocene volcanic and intrusive rocks occur on these islands, and were sampled for this paleomagnetic study. Eocene sandstones were sampled on San Miguel Island, and Eocene sandstones and Miocene siltstones were sampled on Santa Cruz Island. Paleomagnetic results from igneous units are characterized by declinations deflected clockwise by 69° to 81° from expected directions and inclinations which are too shallow by 10° to 25°. Normal and reversely magnetized units yield antipodal mean directions. The mean result is concordant to the directions obtained from individual units which passed fold and baked contact tests. These results suggest that the northern Channel Islands have been tectonically rotated into place since early(?) Miocene time as the outer borderland area translated northwestward within a large shear zone between the Pacific and North American Plates. The data from the northern Channel Islands when averaged with the data from the Santa Monica Mountains yield a result of $I = 36.1° \pm 5.1°$, $D = 72.6° \pm 6.3°$ for the southern part of the western Transverse Ranges (52 units, 405 samples). The shallow paleomagnetic inclinations suggest a northward latitude translation of $14.0° + 3.7°$ or $-3.9°$ for the western Transverse Ranges. However, most plate tectonic reconstructions suggest only $4° \pm 3°$ of northward translation. The discrepancy may be due to initial dips of the volcanic flows which would cause an artificial shallowing of inclination when the structural dip of the units was corrected to an assumed original horizontal. Other causes for the shallow inclinations could be the effects of an offset dipole field or nondipole field components, or motions of small tectonic plates between the North American and Pacific plates as the Farallan plate broke up and was subducted.

INTRODUCTION

The northern Channel Islands of California lie along the southern margin of the western Transverse Range physiographic structural province (Figure 1). The western Transverse Ranges are characterized by east-west topographic, structural, and lithologic trends, in contrast to the typical northwest-southeast trend of these features elsewhere in California. Based on analysis of clay-cake experiments and fracture patterns, *Cloos* [1955] suggested that the present fault pattern of southern California implies a large amount of clockwise rotation of transverse fault-bounded blocks. The anomalous trends of fold axes, structures, and lithic belts within the Transverse Ranges lead *Jones and Irwin* [1975] to suggest that the western Transverse Ranges were tectonically rotated into their present position. *Jones et al.* [1976] further suggested that the rotation was clockwise. *Baird et al.* [1974] considered rotation of the Transverse Ranges but concluded that the amount of rotation necessary to bring the trends within the Transverse Ranges into realignment with correlative features in the rest of California was untenable.

MacKinnon [1976] measured the paleomagnetism of Cretaceous rocks in the San Rafael Mountains, and proposed a large clockwise rotation of that area. However, these data are inconclusive since the samples are all normally magnetized and no field tests were performed. *Kamerling and Luyendyk* [1977, 1979*a*] and *Kamerling* [1980] studied the paleomagnetism of middle Miocene volcanic rocks in the Santa Monica Mountains and on Anacapa Island. Their results indicated that the paleomagnetic declination was deflected clockwise from the expected direction by 75° and that the inclination was too shallow by about 10°. They concluded from these results that the western Transverse Ranges had been rotated clockwise at least 75° since middle Miocene time as a microplate or structural block within the northwest trending shear couple between the Pacific and North American plates. Based on the trends of lithic belts in the southern California continental borderland, *Crouch* [1979] suggested that the western Transverse Ranges were rifted northwestward from a position off Baja California and rotated into place when the northern end encountered a westward bulge in the continent at the southern end of the Coast Ranges.

Hall [1978, 1981*a*, *b*] suggested that the western Transverse Ranges have rotated counterclockwise into place, thus opening up the Santa Maria basin. *Howell* [1976*a*] also proposed a counterclockwise rotation of the southern part of Santa Cruz Island with respect to the northern part based on the structural trends within the Santa Cruz Island Schist and the Santa Catalina Island Schist.

The present study was undertaken to extend the area of the earlier work of *Kamerling and Luyendyk* [1977, 1979*a*] and to provide constraints on these models and others concerning the tectonic evolution of the borderland.

PROCEDURE

Altogether, 459 individually oriented samples were collected from Santa Cruz, Santa Rosa, and San Miguel Islands. The samples (420) are mainly from latest Oligocene to middle Miocene age volcanic and intrusive rocks of the San Miguel Volcanics and Santa Cruz Island Volcanics [*Weaver and Doerner*, 1969; *Nolf and Nolf*, 1969].

Middle Miocene sedimentary rocks were collected (20 sam-

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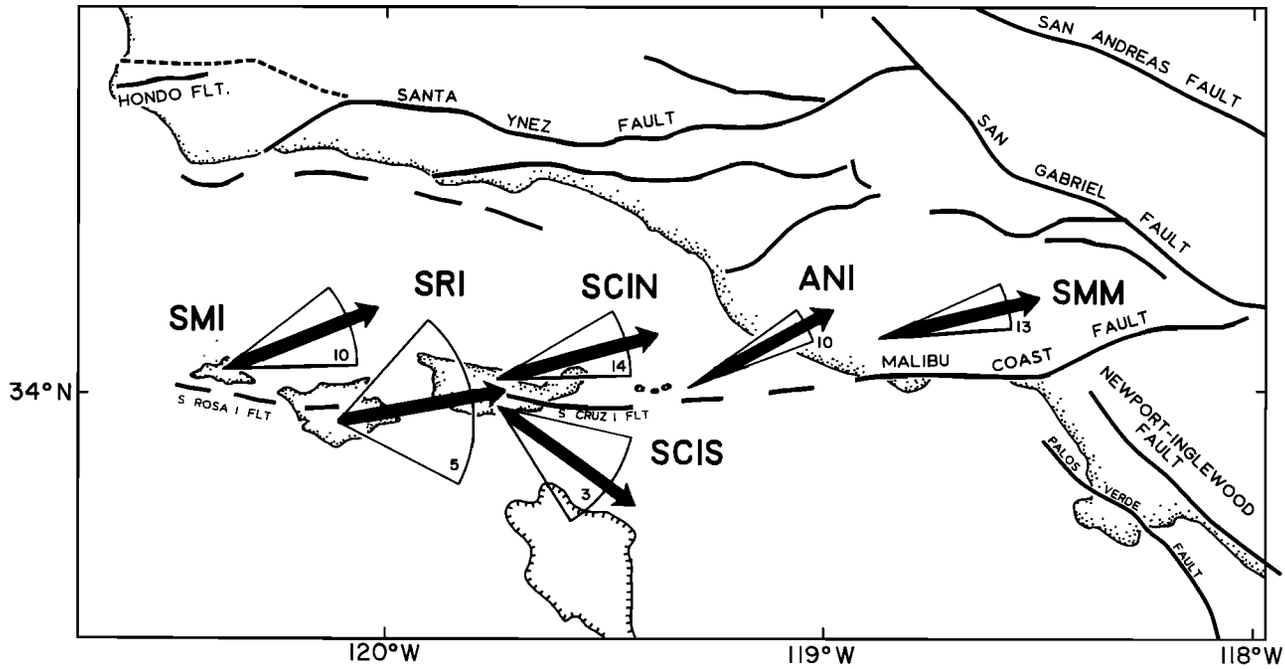


Fig. 1. Location map of the northern Channel Islands in the western Transverse Ranges province; mean declinations and associated 95% confidence limits on the declination are shown ($\Delta D = \sin^{-1}$ (sin alpha 95/cos I). The results for SMM and ANI are from *Kamerling and Luyendyk* [1979*a*]. Numbers next to the declination arrows indicate the number of units averaged for the mean. SMI is San Miguel Island, SRI Santa Rosa Island, SCI Santa Cruz, ANI Anacapa, and SMM the Santa Monica Mountains.

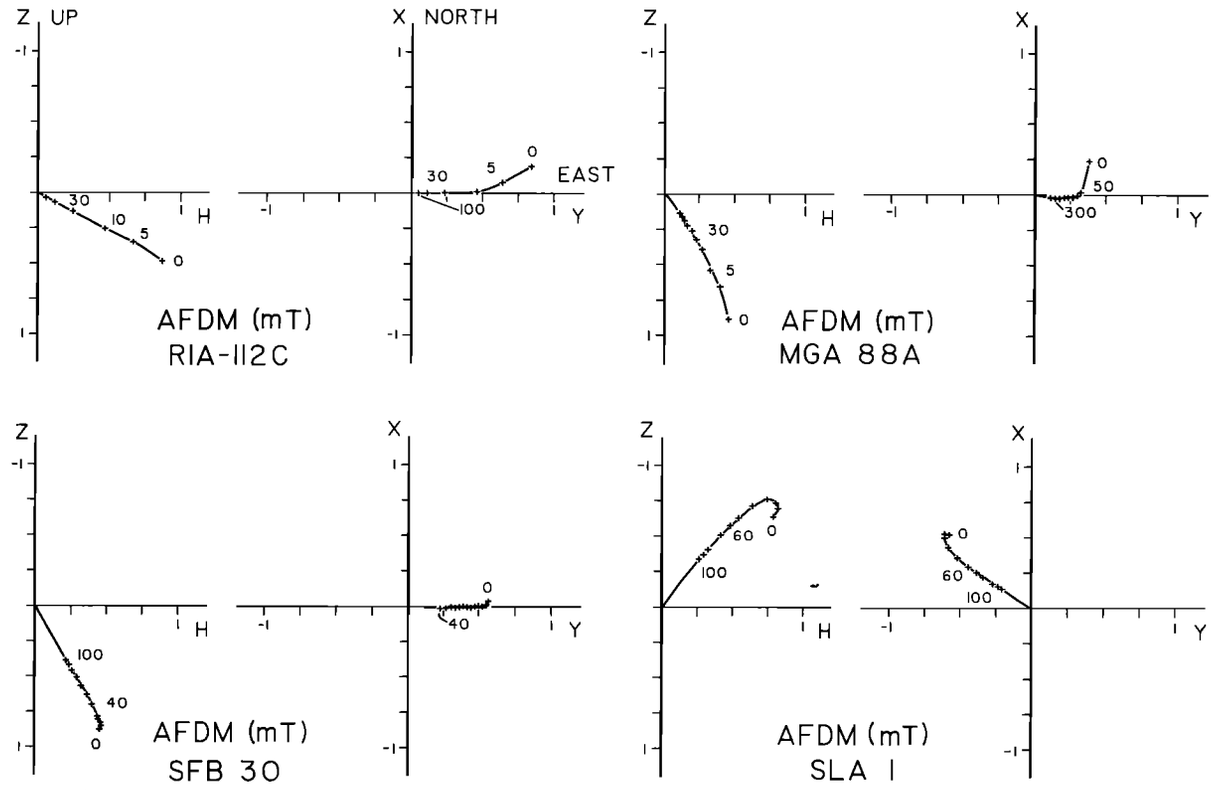


Fig. 2. Vector end point or *Zijderveld* [1967] plots of alternating field demagnetization series for rocks from the northern Channel Islands. Sample RIA is from a basaltic intrusive on south Santa Rosa Island (Figure 5); MGA from a lava flow on south San Miguel Island (Figure 7); SFB from a flow on north Santa Cruz Island (Figure 3); and SLA from a sill on south Santa Cruz Island (SCI, Figure 3). Units of demagnetization steps are in milliteslas (mT).

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TABLE 1. Santa Cruz Island Uncorrected and Corrected Directions

Unit	Member*	Type	N	I_u†, deg	D_u†, deg	I, deg	D, deg	α_{95}	af[3] Range‡, mT	T, °C§
				North Santa Cruz Island						
SAA	G	FLOW	7	06.1	68.0	06.1	068.0	3.1	20–70	
SAB	G	FLOW	5	11.8	71.9	11.8	071.9	3.3	20–60	
SBB	G	FLOW	5	49.8	95.2	53.7	060.2	3.3	30–80	
SDA	G	FLOW	7	13.3	73.2	23.2	072.9	4.3	30–60	
SFB	G	FLOW	6	61.4	95.1	57.5	079.4	2.1	40–70	
SFC	G	FLOW	7	67.1	91.3	62.3	072.5	6.0	40–80	
SCFA	D	FLOW	6	04.9	86.6	11.8	091.8	4.1	30–70	
SCVA	D	FLOW	9	−71.3	257.1	−52.2	241.7	3.8	10–40	
SCVB	D	FLOW	7	−15.6	270.9	−09.7	267.7	3.7	40–100	
SCVC	D	FLOW	6	−27.4	267.9	−16.4	259.0	11.2	60–100	
SCEA	D	FLOW	11	02.9	239.8	−45.4	241.8	6.1	20–100	
SPA	P	FLOW	10	−37.2	282.0	−22.5	280.3	8.2	30–100	
SSA	P	FLOW	5	29.7	283.8	−27.6	283.7	23.2	80–100	
SSB¶	P	FLOW	6	31.5	006.5	17.2	338.2	1.6	20–90	
SFA	P	DIKE	6	21.8	049.1	13.1	047.6	4.5	50–90	
				South Santa Cruz Island						
SJVA¶	E	SDSTN	5	−17.5	150.7	16.3	126.6	22.7	40–80	
SGA	B	SILL	9	−24.6	308.5	−14.8	312.7	6.5		350
SCLA	B	SILL	15	−48.6	295.8	−30.1	299.3	2.8	15–100	
SWAA	B	DIKE	10	−64.0	222.4	−36.5	304.8	2.4	40–80	
SQAA¶	B	DIKE	3	54.8	222.0	−03.2	221.2	11.0	25–40	

Paleomagnetic data for units from Santa Cruz Island. The results are divided into those from the northern and the southern parts of the island. For the results from Anacapa Island and the Santa Monica Mountains, see *Kamerling and Luyendyk* [1979*a*]. N, number of units used to calculate mean direction; I, inclination; D, declination; α_{95}, radius of the cone of 95% confidence.

*Members of Santa Cruz Island volcanics and their ages (corrected); Blanca Formation, B, 14.9 ± 0.8 Ma [*McLean et al.*, 1976*a*]; Prisoners Harbor, P, 16.5 ± 0.8 Ma and 16.1 ± 0.9 Ma [*Turner*, 1970]; Devils Peak, D, 15.9 ± 0.7 Ma and 19.9 ± 0.9 Ma (plag, same flow [*Crowe et al.*, 1976]); Griffith Canyon, G, middle Miocene; Eocene marine, E.

†In situ directions, uncorrected for tilt.

‡Demagnetizing field where directions stabilized.

§Demagnetizing temperature where directions stabilized.

¶Not used in averages

ples) from the southern part of Santa Cruz Island and Eocene sandstones were sampled from Santa Cruz (seven samples) and San Miguel Islands (12 samples). All of the samples were cored in the field with a portable drill, except for 13 which were collected as oriented blocks. Orientations were determined by Brunton compass and checked with sun sights or bearings to landmarks. Flows were corrected to an assumed original horizontal by determining the attitude of the flow and interbedded sediments if available. Intrusive units sampled were corrected by restoring the sediments which they intruded to horizontal about the strike of the beds. Care was taken to measure attitudes away from any disturbance caused by intrusion.

Each sample was individually stepwise demagnetized using either alternating field or thermal demagnetization, or sometimes both if more than one 2.5 cm long specimen was available. Stable directions were picked with the aid of Zidjerveld plots [*Zidjerveld*, 1967] (Figure 2), direction vector differences between demagnetization levels, and equal area plots. The directions of individual samples (an average if two or more specimens from the same sample were available) were averaged to determine the vector mean of the cooling unit (flow, dike, or sill) from which they were collected (Tables 1–3). We attempted to take seven to nine samples from each unit by sampling at several locations separated by at least a few to hundreds of meters. However, this was not always possible. Unit means were combined to calculate vector means for each island and for all of the islands (Table 4).

The method of *Beck* [1980] was used to compare the results of this study to reference directions from the stable portion of North America in order to establish rotation and flattening anomalies. *Demarest* [1983] has determined that true confidence limits on the anomalies are 75% to 80% of the values determined by Beck's method. Therefore, the correction factors determined by *Demarest* [1983] have been applied to confidence limits for direction anomalies with respect to reference directions. The uncertainties of individual or mean results are not affected by this correction.

All K-Ar ages quoted in the text have been corrected where necessary, to the new decay constants using the formula given by *Ness et al.* [1980].

PALEOMAGNETIC RESULTS

Santa Cruz Island

Santa Cruz Island is the largest of the Channel Islands off southern California. The island is cut into northern and southern parts by the east-west trending Santa Cruz Island fault (Figure 3) which forms a pronounced central valley and offsets stream patterns [*Weaver*, 1969]. *Luyendyk et al.* [1983] suggest 10 km of post Miocene left-slip based on an apparent offset of the Christi anticline on southwest Santa Cruz Island.

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TABLE 2. Santa Rosa Island Uncorrected and Corrected Directions

Unit	Member*	Type	N	I_u†, deg	D_u†, deg	I, deg	D, deg	α_{95}	af[3] Range‡, mT	T, °C§
RAA	SMV	INTRU	8	33.7	049.8	49.8	059.5	6.6	30–50	300
RBA	SMV	SILL	8	51.1	051.8	36.8	038.0	2.5	20–40	300
RCA	SMV	INTRU	9	67.4	021.1	33.8	014.0	2.5	25–60	400
RGA	SMV	DIKE	12	38.8	082.4	38.9	067.7	5.2	25–40	
RIA	SMV	INTRU	8	37.6	100.0	33.6	118.0	7.7	20–40	300
RJA	SMV	INTRU	6	32.1	092.0	44.6	109.6	7.3	30–40	
RKA††	BB	TUFF	7	53.7	033.3	40.8	039.4	5.7	15–40	
RKB	BB	TUFF	7	48.0	012.5	42.3	023.2	5.1	30–40	
RHA††	BB	TUFF	6	40.8	034.5	34.7	034.5	3.8	15–50	
RHB	BB	TUFF	6	42.4	036.3	36.4	036.2	5.4	20–40	
RHC	BB	TUFF	4	40.2	030.1	34.2	030.5	4.3	25–35	
RHD	BB	TUFF	5	35.7	036.1	29.7	035.9	5.5	15–30	
RHE	BB	TUFF	5	27.5	024.1	21.6	024.6	6.3	15–30	
RHF	BB	TUFF	4	39.7	047.1	33.8	046.2	3.9	10–30	
RHG	BB	TUFF	5	41.7	039.5	35.7	039.1	3.7	10–30	
MEAN¶	SMV		6			46.0	067.0	27.3		
MEAN‖	SMV		5			45.0	078.9	26.0		
MEAN**	SMV		4			44.6	090.1	26.5		

Paleomagnetic data for units from Santa Rosa Island. Symbols are as in Table 1. RH and RK units are tuff beds. Means for the intrusive units include all (6), or exclude unit RCA (5), or units RCA and RBA (4).

*BB, Beechers Bay tuff member of Monterey Formation, middle Miocene; SMV, San Miguel Volcanics, early Miocene.

†In situ directions uncorrected for tilt.

‡Demagnetizing field where directions stabilized.

§Demagnetizing temperature where directions stabilized.

¶All intrusive units of SMV; corrected directions.

‖Excluding RCA.

**Excluding RCA and RBA.

††Units RH and RK are believed to be late Miocene in age; the other intrusive units are possibly late Oligocene to early Miocene if they correlate with the San Miguel Island Volcanics.

The fault separates the island into distinct stratigraphic sequences. On the north side of the island only Miocene volcanic rocks and Miocene through Holocene sedimentary rocks are exposed. South of the Santa Cruz Island fault a Jurassic basement complex of Santa Cruz Island Schist intruded by the Alamos leucotonalite and faulted against the Jurassic Willows Diorite, is in turn faulted against a Paleocene through Miocene sedimentary and volcaniclastic rock sequence [*Doerner*, 1969; *Bereskin and Edwards*, 1969; *Hill*, 1976]. The difference between the stratigraphic sequences north and south of the Santa Cruz Island fault extends up through the middle Miocene Upper Blanca Formation, although clasts in the upper part of the Blanca Formation resemble the andesitic Santa Cruz Island Volcanics [*McLean et*

TABLE 3. San Miguel Island Uncorrected and Corrected Directions

Unit	Type	K-Ar Age, Ma Corrected	N	I_u*, deg	D_u*, deg	I, deg	D, deg	α_{95}	af† Range, mT	T, °C‡
MA	INTRU	25.0 ± 3.0	9	−37.5	284.1	−29.9	275.4	7.6	30–80	
MBA	FLOW		6	52.6	40.4	37.7	054.2	4.4	40–80	450
MBA	FLOW	28.6 ± 3.2	8	36.5	62.2	10.1	055.8	2.2	60–80	370
MBC	FLOW		4	39.2	57.1	11.6	051.1	8.5	20–30	370
MBD	DIKE		5	39.4	62.9	13.0	055.5	8.4	30–60	
MCA	DIKE		7	47.4	81.0	17.7	069.7	5.9	15–25	450–525
MCB	DIKE		12	58.2	123.9	46.4	094.2	4.5	20–80	
MDA	FLOW		9	59.2	125.1	44.0	108.1	5.2	30–50	450
MEA§	FLOW		8	32.1	34.2	40.6	026.9	3.7	40–60	450
MGA	FLOW	31.8 ± 3.5	10	59.8	101.2	37.7	083.4	1.8	40–50	450
MGB	FLOW		5	64.4	22.5	42.2	044.5	5.8	30–60	450

Paleomagnetic data for units from San Miguel Island. Symbols are as in Table 1. The mean result excludes unit MEA. MA is from the upper member of the San Miguel volcanics, and the rest of the units are from the lower member of the San Miguel volcanics.

*In situ direction, uncorrected for tilt.

†Demagnetizing field where directions stabilized.

‡Demagnetizing temperature where directions stabilized.

§MEA not used to calculate mean direction.

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TABLE 4a. Average Directions and Paleomagnetic Poles for the Locations Studied (Corrected for Tilt of Beds)

Location	N	I, deg	D, deg	Site Latitude, deg	Site Longitude, deg	α_{95}	k	VGP Latitude, deg	VGP Longitude, deg	Age, Ma
SMM*	13	45.0	074.7	34.1	−118.9	6.3	43.7	26.46	−044.31	13–16 ± 1
ANI*	10	32.4	064.0	34.0	−119.4	5.3	85.5	31.01	−030.37	12–17
SCIN	14	30.3	076.9	34.0	−119.8	12.5	11.2	19.73	−036.56	16–20 ± 1
SCIS	3	27.3	125.9	34.0	−119.8	19.6	40.8	19.35	116.46	13–15 ± 1
SRII	5	45.1	078.9	34.0	−120.1	26.0	9.6	23.18	−047.63	16–34
SRIT	9	34.6	034.4	34.0	−120.1	5.5	90.0	56.02	−012.76	11.5–16.5
SMI	10	30.7	069.6	34.0	−120.3	14.6	12.0	25.88	−033.39	25–32 ± 3
WTR†	52	36.1	072.6	34.0	−119.6	5.1	15.8	25.15	−037.64	14–32 ± 3
WTR(N)	27	35.6	071.4	34.0	−119.6	8.2	12.5	25.95	−036.74	14–32 ± 3
WTR(R)	25	−36.5	253.4	34.0	−119.6	6.3	21.9	−24.58	141.68	14–25 ± 3

SMM, Santa Monica Mountains; ANI, Anacapa Island (both from *Kamerling and Luyendyk* [1979*a*]; SCIN, northern part of Santa Cruz Island; SCIS, the southern part of Santa Cruz Island; SRII, Santa Rosa Island intrusive units; SRIT, Santa Rosa Island tuffs; SMI, San Miguel Island; WRT, western Transverse Ranges mean which includes the data of *Kamerling and Luyendyk* [1979*a*] but excludes South Santa Cruz Island; WTR(N), western Transverse Ranges normally magnetized data; WTR(R), western Transverse Ranges reversely magnetized data. Age is the range in K-Ar dates of the rocks sampled with the associated precision of the K-Ar dates. Where only a range is given, this is an estimated age based on correlations.

*From *Kamerling and Luyendyk* [1979*a*].

†WTR average excludes units from South Santa Cruz Island and tuff beds on Santa Rosa Island.

al., 1976*a*; *Fisher and Charlton*, 1976]. Preliminary data of *Kamerling et al.* [1978, 1980] showed a declination difference of about 50° across the fault, which when considered along with the stratigraphic differences, supports the suggestion of *Howell* [1976*a*] that the Santa Cruz Island fault is a major tectonic break. However, the large difference in declinations could be due to an excursion in the geomagnetic field since the units measured south of the fault are all reversely magnetized and are fairly close in paleomagnetic direction.

The rocks we sampled on the north side of the island are of the Santa Cruz Island Volcanics [*Nolf and Nolf*, 1969]. This volcanic sequence, up to 2440 m thick, is divided into four members based on lithology. These members are from bottom to top: The Griffith Canyon, Stanton Ranch, Devil's Peak, and Prisoners Harbor members. Of the 17 units (134 cores) sampled on the northern part of the island, one unit proved to be unstably magnetized, one was partially remagnetized by the flow directly above, and another unit (SBA) yielded a highly anomalous direction. Of the remaining units, six were from the Griffith Canyon member, five were from the Devil's Peak member, and one unit was from the Prisoners Harbor member. One flow and two dikes were from rocks classified as Santa Cruz Island Volcanics, undifferentiated, by *Weaver* [1969], and *Nolf and Nolf* [1969]. The Stanton Ranch member was not sampled because the rocks are mainly breccias and agglomerates. The uppermost Prisoners Harbor member has been dated in two locations by K-Ar methods, which yielded dates of 16.5 ± 0.8 and 16.1 ± 0.9 Ma (corrected ages of *Turner* [1970]). A dike intruding the Devil's Peak member yielded a whole rock K-Ar age of 15.9 ± 0.7 Ma and a plagioclase separate from the same rock gave an age of 19.9 ± 0.9 Ma (corrected ages; reported by *Crowe et al.* [1976]). All six of the units sampled from the Griffith Canyon member are of normal polarity while all but one unit from the Devil's Peak member and the Prisoners Harbor member are reversely magnetized. The age of the reversed Prisoners Harbor member fits the geomagnetic reversal time scale [*Ness et al.*, 1980], and the whole sequence may fit within magnetic epoch 16, anomaly 5C. The average result from 14 units (96 samples) on the northern side of Santa Cruz Island is 30.3° ± 12.5° inclination, 76.5° ± 14.5° declination (Figures 1, 4, and 10; Table 4).

South of the Santa Cruz Island fault, 91 samples were collected from 17 units. These include two sills intruding the upper member of the Blanca Formation, 2 dikes intruding the middle member of the Blanca Formation, tuff, tuffaceous sandstone, and sandstone of the middle and upper members of the Blanca Formation, and siltstone of the Beachers Bay member [*Bereskin and Edwards*, 1969] of the Monterey Formation. K-Ar dates from a basaltic flow in the upper Blanca Formation, and from a dacite clast from the upper part of the formation yield ages of 14.9 ± 0.8 Ma and 13.3 ± 1.2 Ma, respectively (corrected ages; reported by *McLean et al.* [1976*b*]). This implies that the units sampled which intrude

TABLE 4b. Average Directions for Units Uncorrected for Tilt of Beds

Location	I_w, deg	D_w, deg	α_{95}	k
SMM*	37.2	94.6	10.2	17.4
ANI*	39.0	77.8	5.2	86.4
SCIN	26.2	80.7	16.9	6.5
SCIS	52.8	157.8	63.1	4.9
SRII	40.5	76.3	18.6	17.8
SRIT	41.4	32.8	6.6	62.4
SMI	51.2	69.3	15.4	10.8
WTR†	38.6	82.4	6.1	11.6

Same islands and units as in Table 4a; however, all means are computed for in situ paleomagnetic directions, uncorrected for tilt and plunge of folded beds.

*From *Kamerling and Luyendyk* [1979*a*].

†WTR average excludes units from South Santa Cruz Island and tuff beds on Santa Rosa Island.

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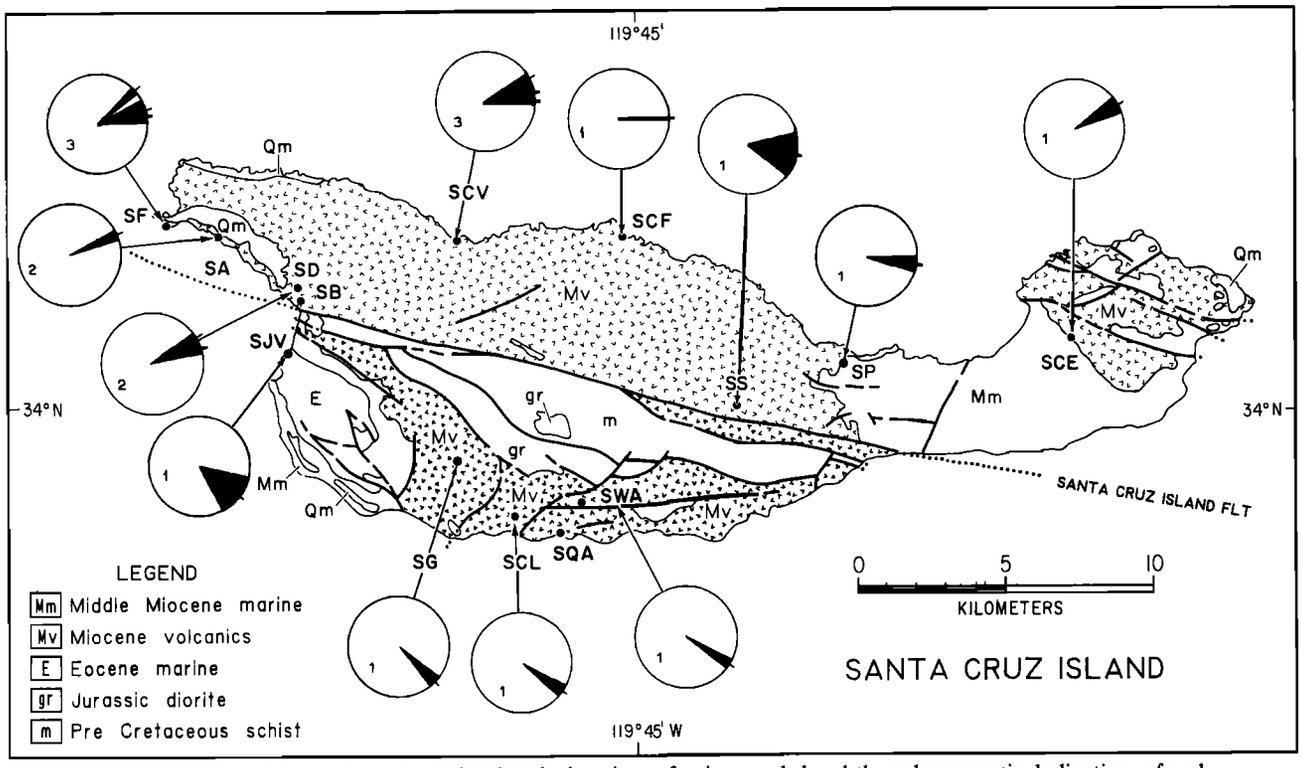


Fig. 3. Map of Santa Cruz Island showing the locations of units sampled and the paleomagnetic declinations of each unit with the associated 95% confidence limits. Numbers within circles indicate the number of units at each location. No stable results could be found at location SQA.

the upper member of the Blanca Formation are younger than the Santa Cruz Island Volcanics north of the Santa Cruz Island fault. Also, seven Eocene sandstone samples were collected from the Jolla Vieja Formation on the western coast of the island by J. S. Hornafius for the authors (SJV in Figure 3).

The Miocene sedimentary samples proved to be unstably magnetized or yielded widely scattered results. One of the dikes sampled (SQA) also yielded poor magnetic data and a highly divergent direction. The three remaining units (34 samples from SGA, SCLA, and SWA) were all of reverse polarity and yield a mean result (inverted to equivalent normal polarity) of 27.3° ± 19.6° inclination, 125.9 ± 22.2° declination

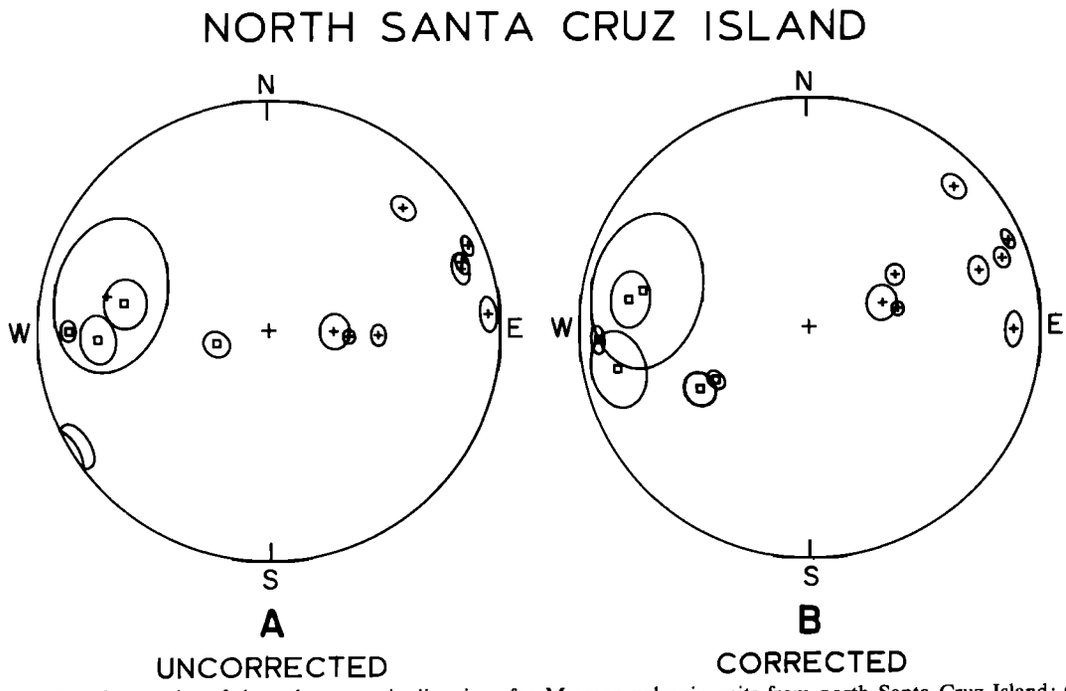


Fig. 4. Equal-area plot of the paleomagnetic directions for Miocene volcanic units from north Santa Cruz Island: (*a*) before and (*b*) after structural corrections.

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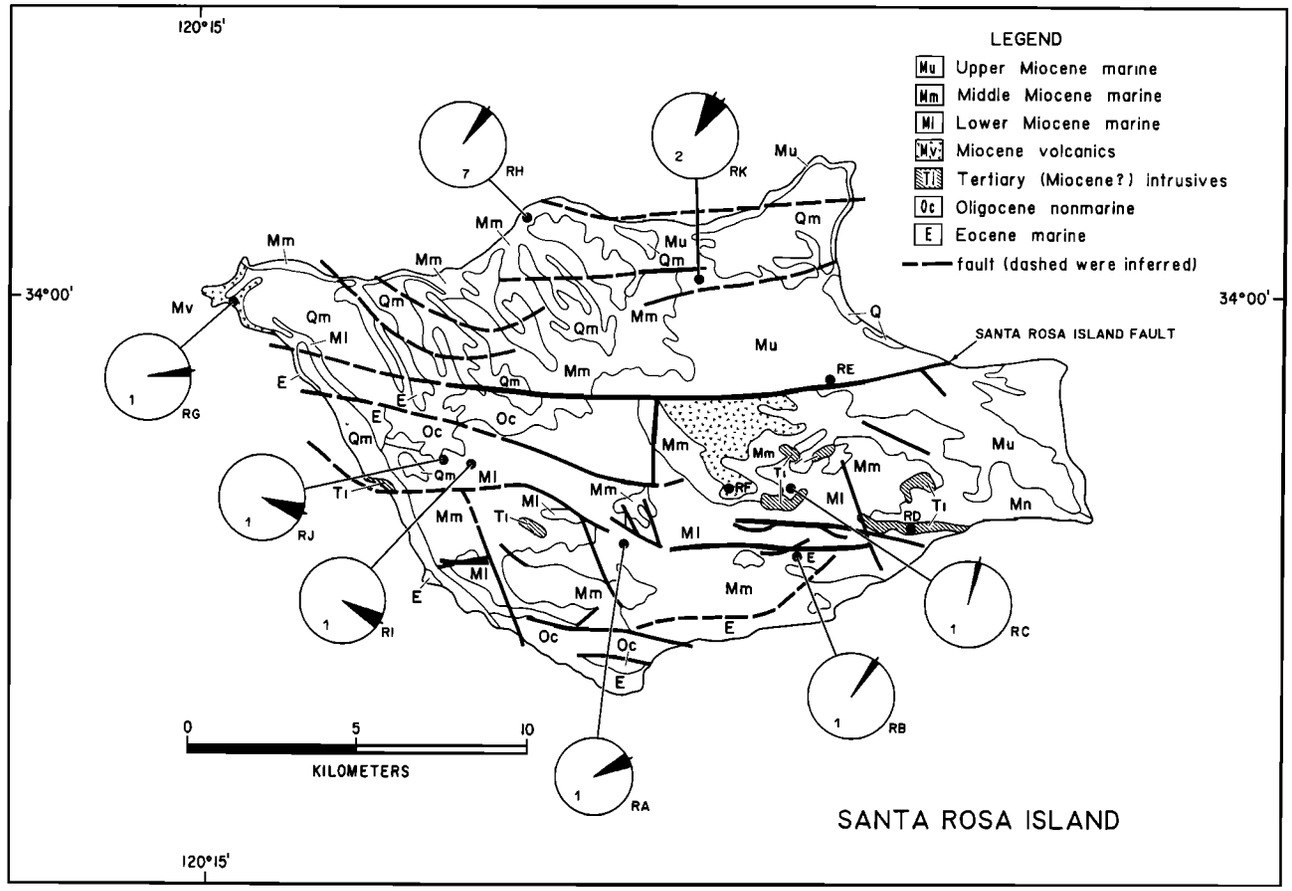


Fig. 5. Map of Santa Rosa Island showing the locations of the units sampled and the paleomagnetic declinations for individual units or locality averages (RH). No stable result could be obtained at locations RD, RE, and RF.

(Figure 10, Table 4). Although secular variation is obviously not averaged in such a small sample, the result is deflected 48° clockwise from the mean result for the northern part of the island, which is more than can typically be attributed to secular variation. Also, since the units are widely distributed it seems unlikely that the result could be due to rotation of local faulted blocks. The fact that all three units are reversely magnetized and so close in direction suggests the possibility that they represent one intrusive episode. If this is the case this result could be due to an excursion of the geomagnetic field. The seven Eocene sedimentary rock samples collected were unstable to alternating field demagnetization. However, with thermal demagnetization the sample directions clustered somewhat at 350°C, and after correcting for plunge and tilt of the beds, 5 samples gave a mean result of 16.3° inclination, 126.6° declination, $\alpha_{95} = 22.7°$. This result has large error limits, but it shows large clockwise rotation and inclination flattening and enhances the credibility of the result from the Miocene volcanic rocks.

There does not appear to be a clear association of declination change with distance from the Santa Cruz Island fault as was found by *Terres et al.* [1980] adjacent to the right-lateral Clemens Well fault northeast of the Salton Sea. Therefore, the scatter of declinations on Santa Cruz Island is not due to distributed shear adjacent to a strike-slip fault.

Santa Rosa Island

Santa Rosa Island lies directly to the west of Santa Cruz Island and like Santa Cruz Island it is cut by a large east-west trending median fault (Figure 5). The Santa Rosa Island fault apparently abuts against the Santa Cruz Island fault just west of Santa Cruz Island [*Junger*, 1976*a*; *Luyendyk et al.*, 1983]. From this point the fault traces a gentle curve southwest through the center of Santa Rosa Island continuing westward to south of San Miguel Island. The fault then makes a bend to the northwest and is lost northwest of San Miguel Island [*Junger*, 1979]. *Anderson et al.* [1949] have suggested a lateral offset of 8 km on the Santa Rosa Island fault, and *Weaver* [1969] suggested about 16 km of left-lateral displacement based on offset facies of the San Miguel volcanics.

Although the structural style changes dramatically on either side of the fault (Figure 5), the stratigraphic sequence exposed and in the subsurface appears to be the same [*Anderson et al.*, 1949; *Weaver and Doerner*, 1969; *Avila and Weaver*, 1969]. South of the Santa Rosa Island fault, Eocene sandstone, Oligocene nonmarine (Sespe Formation), and Miocene marine beds are exposed in complex fault wedges and folds. These sedimentary rocks are intruded by basaltic sills, dikes and irregular-shaped shallow intrusions of the San Miguel Volcanics. Based on stratigraphic and intrusive relationships these

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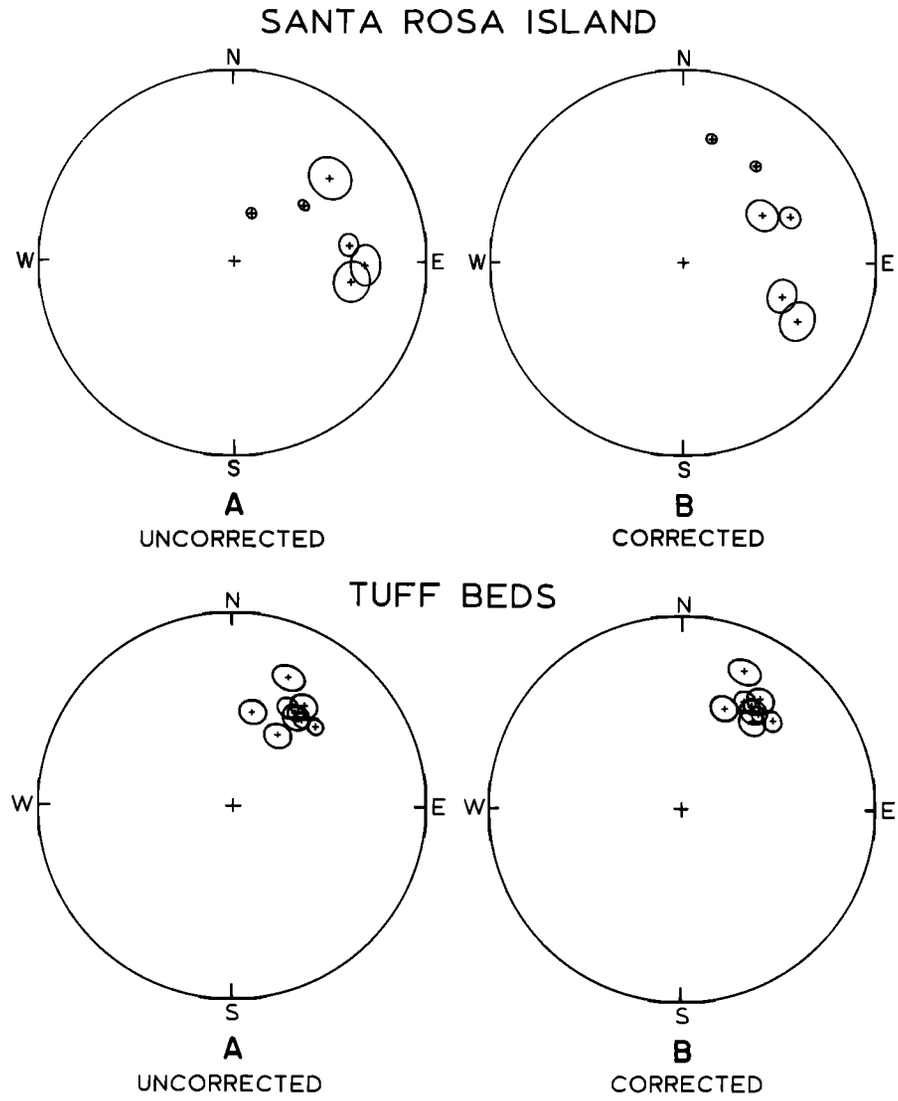


Fig. 6. Equal-area plot of the paleomagnetic directions from the intrusive volcanic units (upper) and tuff beds (lower) from Santa Rosa Island: (*a*) before and (*b*) after structural corrections.

volcanic rocks are late Oligocene to early Miocene in age. The intrusive rocks occur mainly south of the Santa Rosa Island fault. Only one dike occurs north of the fault on the northwest coast of the island. Gently dipping beds of the Vaqueros Formation, Rincon Shale, San Onofre Breccia, and the Beechers Bay Member of the Monterey Formation [*Avila and Weaver*, 1969] are also exposed north of the fault. *McLean et al.* [1976*b*] consider strata described as San Onofre Breccia and Beechers Bay Member of the Monterey Formation by *Avila and Weaver* [1969] to be a facies of the Blanca Formation.

Mostly intrusive rocks of the San Miguel Volcanics were sampled on this island. K-Ar dates are not available for this formation on the island, but it is assumed here to be of early Miocene age. The paleomagnetic results are from three sills, one dike, and two irregular-shaped shallow intrusions (Figure 6). In addition, nine tuff beds of the middle Miocene Beechers Bay Member of the Monterey Formation (or Blanca Formation) were sampled (locations RH and RK, Figure 4 and Table 4). The uncertainties in the structural corrections and age control, and the fact that no reversed units were encountered make the result from Santa Rosa Island weaker than those from the other islands. Also, several of the units exhibited a strong component of magnetization in the present field direction which might not have been completely removed during alternating field or thermal demagnetization (e.g., RCA and possibly RBA). Despite these deficiencies the results are still concordant with data from the other northern Channel Islands in showing a large clockwise deflection in declination and a mean inclination shallower than the expected inclination. Also, the fairly narrow range of inclinations leads to some confidence in the validity of the structural corrections. The tuff beds, which are younger than the San Miguel Volcanics, yield a mean direction of $34.6° \pm 5.5°$ inclination, $34.4° \pm 6.7°$ declination. The older San Miguel intrusives (five units) yield a mean result of $45.1° \pm 26.0°$ inclination, $78.9° \pm 38.4°$ declination excluding unit RCA (Tables 4 and 5).

Recent paleomagnetic work on dolomite concretions from the Rincon Formation at a location midway between RH and

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TABLE 5. Rotation and Flattening of Paleomagnetic Directions Compared to Reference Directions

Area	Age	N	I, deg	D, deg	α_{95}	ΔI	ΔD	R, deg	ΔR	F, deg	ΔF	C
SMM	20	13	45.0	074.7	6.3	6.3	8.9	78.2	8.5	9.3	6.4	0.79
ANI	20	10	32.4	064.0	5.3	5.3	6.3	67.5	6.9	21.9	5.8	0.79
SCIN	20	14	30.3	076.9	12.5	12.5	14.5	80.3	11.7	24.0	10.7	0.79
SRII*	25	5	45.1	078.9	26.0	26.0	38.4	82.5	29.8	8.9	20.3	0.77
SRIT	20	9	34.6	034.4	5.5	5.5	6.7	37.9	7.1	19.7	6.1	0.78
SMI	30	10	30.7	069.6	14.6	14.6	17.0	74.3	14.3	23.8	12.2	0.79
WTR	20	52	36.1	072.6	5.1	5.1	6.3	76.1	7.0	18.2	5.7	0.80

Discordance statistics [*Beck*, 1980; *Demarest*, 1983] of the poles in Table 4*a* compared to the stable North American craton poles of *Irving* [1979]. Positive rotation is clockwise and positive flattening indicates that the observed inclination is too shallow. Age is age of comparison pole (below), $\Delta I = \alpha_{95}$, $\Delta D = \sin^{-1}(\sin \alpha_{95}/\cos I)$, R and ΔR are rotation and adjusted uncertainty and F and ΔF are flattening of inclination and adjusted uncertainty from *Beck* [1980], C is the conversion factor from *Demarest* [1983] used to adjust uncertainties. Reference poles [*Irving*, 1979] 20 Ma pole at 87°N, 167°E, $\alpha_{95} = 5°$; 25 Ma pole at 87°N, 162°E, $\alpha_{95} = 4°$; 30 Ma pole at 85°N, 166°E, $\alpha_{95} = 5°$.

*Islands where there are not enough data to average secular variation, $\alpha_{95} > 15°$. The values calculated for these areas should be considered approximate.

RG on the northwest coast shows declinations consistent with the results from the tuff units (E. Weaver, personal communication, 1985).

San Miguel Island

San Miguel Island (Figure 7) is the westernmost of the northern Channel Islands. An Upper Cretaceous through Miocene section is exposed, although many contacts are faults or are concealed beneath Quaternary terrace deposits. The Pozo-Cañada Formation of lower Paleocene to middle Eocene age (Ynezian-Ulatizian), which lies with apparent conformable relations on the Cretaceous rocks, consists of sandstones and shales containing two conglomerate units similar to the conglomerate exposed in the Cañada Formation on the southwest part of Santa Cruz Island. Both of these conglomerate units resemble Poway Conglomerate which occurs in the

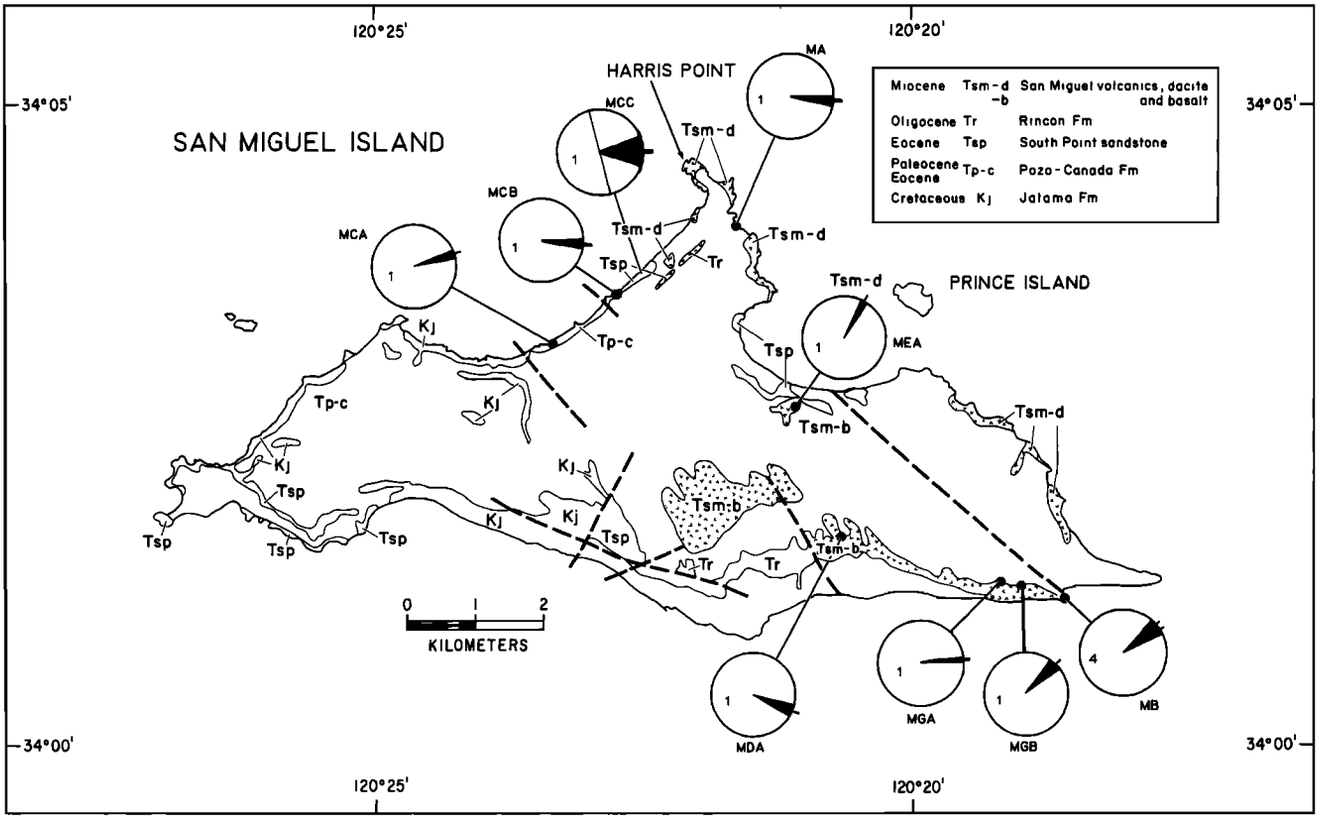


Fig. 7. Map of San Miguel Island showing the locations and paleomagnetic declinations of the volcanic units. Location MCC is Eocene sandstone and the thin line through the direction circle indicates the declination for the Eocene reference pole [*Irving*, 1979].

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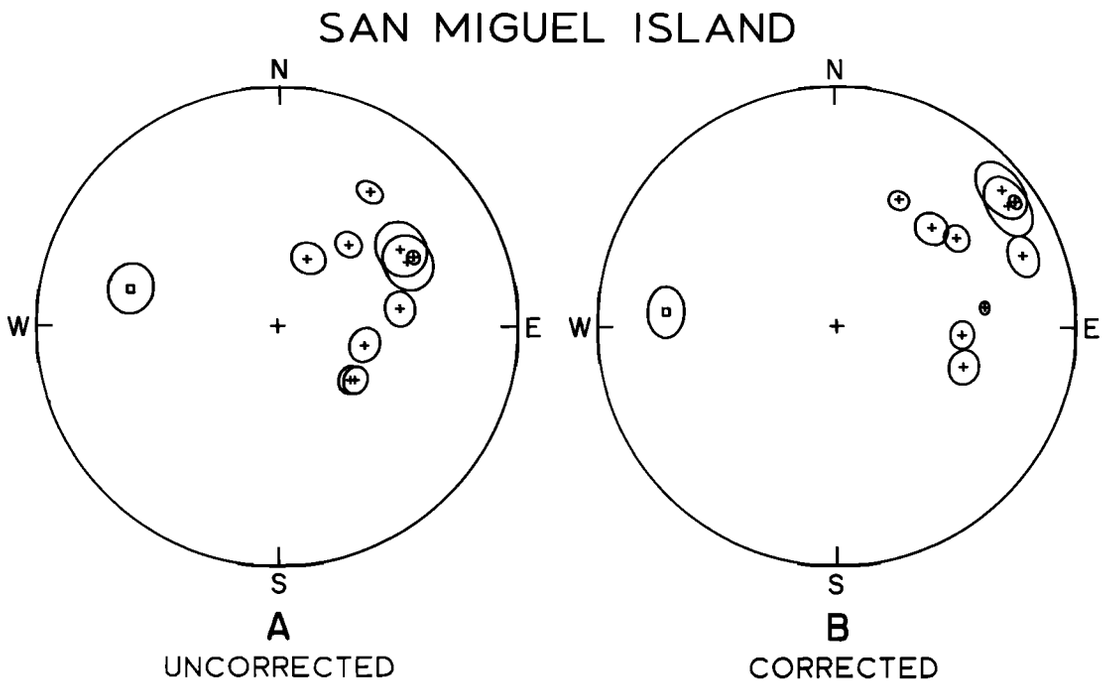


Fig. 8. Equal-area plot of the paleomagnetic directions of late Oligocene (?) to early Miocene volcanics from San Miguel Island: (*a*) before and (*b*) after structural corrections.

San Diego area [*Weaver and Doerner*, 1969; *Cole*, 1970, 1977; *Howell et al.*, 1974; *Erickson*, 1975; *Minch et al.*, 1976; *Abbott and Smith*, 1978; *Kies and Abbott*, 1983; *Abbott et al.*, 1983]. Eocene sandstones of the South Point Formation lie above the Pozo-Cañada Formation which in turn are unconformably overlain by a sequence consisting of the Rincon Formation, San Miguel Volcanics, and Monterey Formation [*Weaver and Doerner*, 1969].

The rocks sampled include the upper and lower members of the San Miguel Volcanics. The lower member comprises volcanic sandstone, volcaniclastics, and flows and dikes which are basaltic in composition. The upper member comprises a dacite plug which outcrops on Harris point and Prince Island and dacitic clastics derived from the plug [*Weaver and Doerner*, 1969]. *Crowe et al.* [1976] report a K-Ar date from dacite of the upper member of the San Miguel Volcanics on Prince Island of 25.0 ± 3 Ma (corrected age). Two K-Ar whole rock dates measured for the authors by Geochron Inc. yield dates of 28.6 ± 3.2 (unit MGA) and 31.8 ± 3.5 Ma (unit MBB). These units are massive flows found on the southeast part of the island (Figure 7) which belong to the lower member of the San Miguel Volcanics. These dates fit the volcanic stratigraphy in light of the date reported for the upper member. However, lower Saucesian Rincon beds apparently underlie the lower member conformably, and clasts from this formation appear in the basal clastics of the lower member of the San Miguel Volcanics [*Weaver and Doerner*, 1969]. The lower boundary of the Saucesian stage is approximately 21 to 22 Ma in age [*Ingle*, 1980]. Therefore, either the K-Ar ages are too old, or the Rincon beds are older than suggested. Although the tectonic implications of the volcanics being as old as the K-Ar ages suggest is important, the discrepancy is of little consequence as far as the paleomagnetic directions are concerned. The expected direction changes by only 1° in declination between the 20 and 30 Ma reference poles [*Irving*, 1979].

The rocks sampled are from seven flows, and three dikes from the lower member of the San Miguel Volcanics, and two sites from the dacite plug on Harris Point (MA on Figure 7) which belongs to the upper member. The dacite plug was structurally corrected for a regional dip which occurs in the Paleogene section to the southwest along Simonton Cove. In addition, 12 samples were collected from the Eocene South Point Formation sandstones (unit MCC, Figure 7) to test the stability of this unit for possible future sampling.

These Oligocene(?) to Miocene volcanic units show the same large clockwise deflections in declination observed elsewhere in the western Transverse Ranges (Figure 8 and Table 4). The inclinations are also as shallow as the data from the other northern Channel Islands. The mean result of 10 units (75 samples) is 30.7° ± 14.6° inclination, 69.6° ± 17.0° declination (Tables 4 and 5). Unit MEA was excluded from the averaging since its direction diverged from the mean by more than 36°. All units but the dacite plug on Harris Point are of normal polarity which is possible supporting evidence for the correlation between these volcanics and those on Santa Rosa Island [*Avila and Weaver*, 1969], which are also all normally magnetized.

Of the 12 Eocene sandstone samples collected from Simonton Cove (unit MCC, Figure 7), only five yielded stable directions. These are concordant with the volcanic directions. One of these stable samples was only 0.5 m away from one of the dikes sampled (part of a baked contact test), and two others were about 8 m away from a 4-m-thick dike which was too weathered to sample. Thus these sandstones may be showing the effects of remagnetization by the volcanic intrusions. However, *Champion et al.* [1981] report a result from Cretaceous through Eocene sedimentary rocks from San Miguel Island which indicate up to 80° of clockwise rotation and northward latitude displacement of 19°, consistent with our results.

A positive fold test (Figure 9) was carried out on a single flow (MGA) which could be traced for about 1.5 km along the

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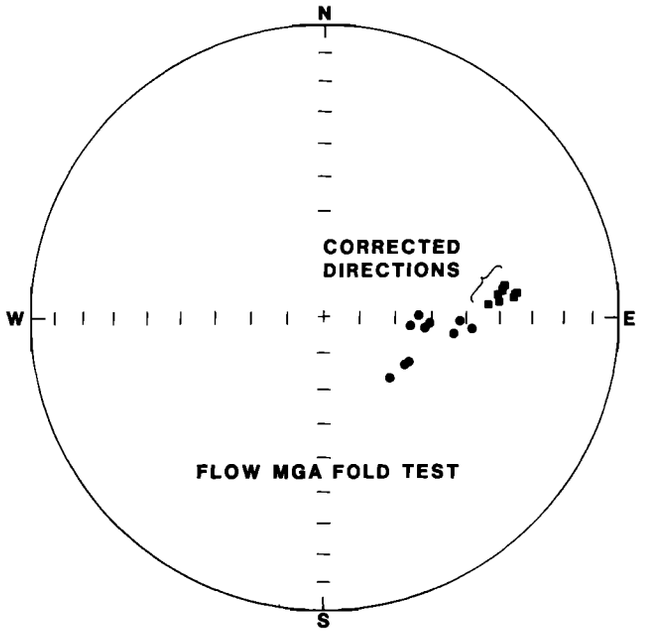


Fig. 9. San Miguel Island unit MGA fold test. Equal-area plot of samples collected in three different locations along a 1.5-km length of flow MGA on San Miguel Island. Both the corrected and uncorrected directions are shown.

southeastern part of the island. Unfortunately, such well-exposed and unweathered units proved to be extremely rare on the island.

Discussion

Including the work of *Kamerling and Luyendyk* [1979*a*] and *Kamerling* [1980] in the Santa Monica Mountains and Anacapa Island, 55 units from Oligocene through middle Miocene volcanics give consistent results from the southern half of the western Transverse Ranges (Figure 10 and Table 4). The four northern Channel Islands yield a mean result of 31.7° ± 7.0° inclination, and 74.4° ± 8.3° declination (33 units excluding the Miocene tuff beds on Santa Rosa Island). If the 13 units from the Santa Monica Mountains are included, [*Kamerling and Luyendyk*, 1979*a*] a result of 35.7° ± 5.6° inclination, 74.5° ± 6.9° declination, is obtained for a total of 46 units. Finally, averaging all units except the three from south Santa Cruz Island and the tuffs from Santa Rosa Island yields $I = 36.1° \pm 5.1°$, $D = 72.6° \pm 6.3°$ for 52 units (Tables 4 and 5).

The 52 units are almost equally divided in polarity (27 normal, 25 reversed; Table 4). Averages were computed separately for the normal and reversed group to apply a reversal test. Table 4*a* and Figure 11 demonstrate that the test is remarkably successful.

The grand mean (WTR in Table 4*a*) may be influenced by the dispersion of declinations, due to local rotation for example. The unit mean directions from Santa Rosa and the Santa Monica Mountains do show more variation with declination than inclination. This could cause the mean inclination to be too steep if it is calculated using *Fisher*'s [1953] statistics; also, alpha 95 could be overestimated for the inclination and underestimated for the declination (R. D. Jarrard, personal communication, 1985). The calculation of inclination only statistics [*Kono*, 1980] gives a grand mean (WTR) inclination of 36.3°, alpha 95 = 5.2° and a precision parameter of 15.1. These values are not very different than the WTR mean values in Table 4*a*, therefore the use of Fisher statistics is not greatly inaccurate for our data.

The number of samples, the consistency of the mean directions, and the wide distribution of sampling sites, lends considerable credibility to the results. The large span of ages of the units sampled (32–14 m.y.) and the fact that the normal and reversely magnetized units are antipodal within about 1°

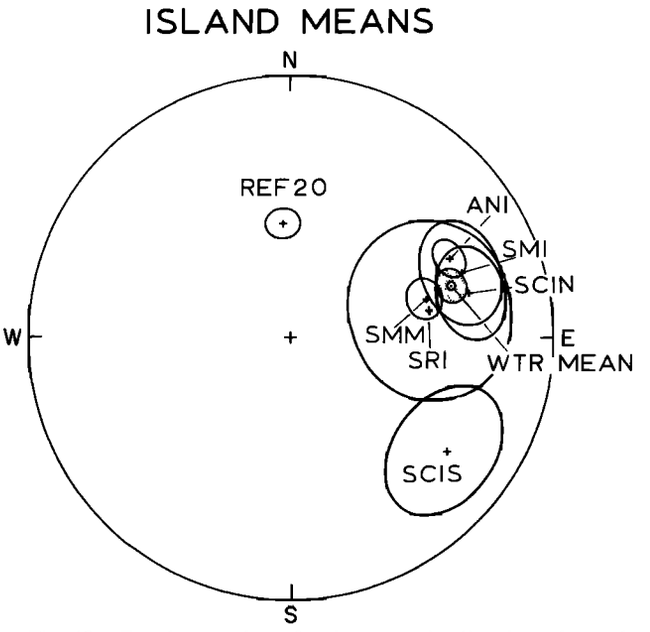


Fig. 10. Equal-area plot of paleomagnetic directions from the northern Channel Islands and the Santa Monica Mountains. The expected direction is indicated as REF 20. The data for East Anacapa Island (ANI) and the Santa Monica Mountains (SMM) are from *Kamerling and Luyendyk* [1979*a*].

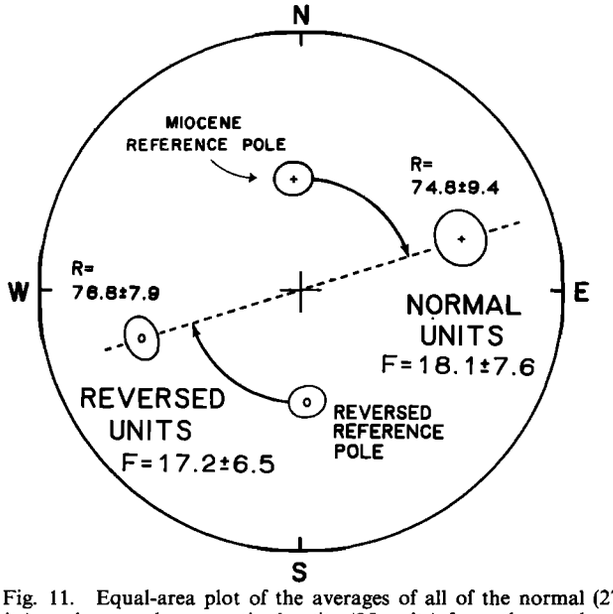


Fig. 11. Equal-area plot of the averages of all of the normal (27 units), and reversely magnetized units (25 units) from the northern Channel Islands and the Santa Monica Mountains, excluding the three units from south Santa Cruz Island and the tuff beds from Santa Rosa Island. The rotations and flattening shown are calculated relative to the 20 m.y. reference pole of *Irving* [1979] using the method of *Beck* [1980].

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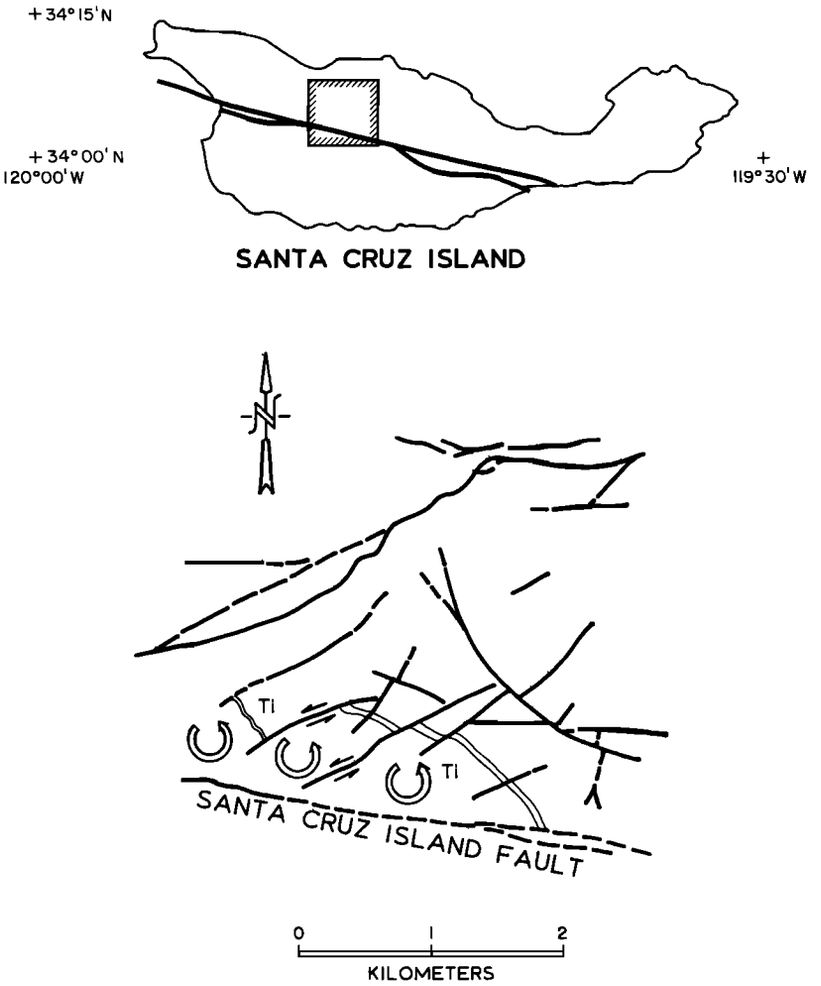


Fig. 12. Map showing inferred counterclockwise rotation of a dike in fault-bounded blocks on the northern part of Santa Cruz Island [from *Weaver*, 1969].

rules out the possibility that the average direction is due to an excursion of field.

The largest uncertainties in this study are due to structural corrections and unknown local rotation of faulted blocks (on the scale of tenths to a few square kilometers). The directions from units on Santa Cruz Island cluster better after structural correction than before, and the San Miguel Island results show little change in declination, though a substantial change in inclination. Directions from Santa Rosa Island increased in dispersion after structural correction (Figure 6 and Table 4). However, this increase is entirely in declination, scatter in inclinations is significantly reduced. The precision parameter $k = 81$ before, and $k = 101$ after structural correction, using the method of *Kono* [1980] for inclination data. The dispersion of declination of the Santa Rosa Island directions is probably an indication of the differential rotation of fault bounded blocks about near vertical axes. Considering the highly faulted nature of the southern part of Santa Rosa Island, this seems a reasonable conclusion.

Typically, paleomagnetic directions from bedded rocks are corrected to a horizontal reference by tilting about the bedding strike by the amount of dip. This procedure makes several assumptions which may not be valid for volcanic flows. Volcanic flows may be deposited with an initial dip of 2° to 12° for basaltic lavas and 5° to 30° for andesitic lavas, and may congeal over very irregular topography. Also, the assumption that the strata were tilted about a horizontal line (the strike line) may not be valid. Structural corrections for intrusive units involve even greater uncertainties since the original relationships of the unit and the strata which it intrudes must be assumed. Structural correction of an intrusive unit by using the attitude of strata which it intrudes or of strata above the unit, is probably a good approach for the northern Channel Islands since most of the deformation is largely late Miocene to Pliocene in age [*Weaver*, 1969] and postdates the intrusives. In addition to the structural correction problems, small blocks may have been rotated about vertical or near vertical axes. Such antithetic rotations can be observed in the offset and rotated fold axes on Santa Rosa Island and in a dike on Santa Cruz Island (Figure 12). The magnitude of these local rotations does not appear to exceed about 30°. The sense of these rotations is ambiguous since the original orientation of the rotated features is unknown. However, *Dibblee* [1950, 1977] has clearly demonstrated counterclockwise rotation of small blocks in the Santa Ynez Mountains which are a part of the Transverse Ranges just north of the northern Channel Islands. The left separation of east-west trending faults and northwest-southeast orientation of folds

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on the islands suggests that the northern Channel Islands area is, or was, within a left-shear couple. This would favor counterclockwise rotation of fault blocks [*Wilcox et al.*, 1973] and reduce the estimate for the maximum amount of clockwise rotation.

MacDonald [1980] suggests a possible method for distinguishing between apparent tectonic rotation caused by tilt corrections incorrectly applied about a horizontal axis and real tectonic rotation. If the paleomagnetic poles for the individual rotated blocks plot along a small circle whose center is near the sampling location, then apparent tectonic rotations are implied. On the other hand, scatter of the pole positions in both azimuth and colatitude, as would be expected for the effects of secular variation, suggests that part of the observed deflection of declination is due to real tectonic rotation. Although the latter is the case for the data from the northern Channel Islands, the unit mean declinations from Santa Rosa Island do seem to be distributed along an arc with little scatter in inclination. This suggests that local rotations are an important effect on the measured declinations from this island.

Another process not considered by *MacDonald* [1980], which would yield an arcuate distribution of pole positions, is formation of the volcanic units through a span of time during the rotation. This possibility was suggested by *Kamerling and Luyendyk* [1979*a*], who observed that progressively younger units in the Santa Monica Mountains showed less apparent rotation. This effect has also been observed in the Monterey Formation [*Hornafius et al.*, 1981, 1982]. However, with the better age control for the Monterey Formation the declinations can be associated with a particular age. With the volcanic rocks sampled for this study, this relationship is more difficult to demonstrate since not enough units for each time interval can be sampled to average secular variation. Also, the age and stratigraphic relationships of some of the units is not known with enough accuracy. Therefore, small magnitude local rotation of fault bounded blocks, and the formation and magnetization of the units during the rotation could be causes of scatter in declination.

The results of this study and the data of *Kamerling and Luyendyk* [1979*a*], when compared to the 20 m.y. reference pole for North America [*Irving*, 1979], indicate an apparent clockwise rotation of 76.1° ± 7.0°, and a mean inclination flattening 18.2° ± 5.7° (Table 5, 52 units). If the factors discussed above have indeed affected the data, then the amount of rotation we calculated should be considered a minimum.

In addition to the rotation anomaly, the data from the western Transverse Ranges exhibit a large inclination anomaly. In fact the data for the western Transverse Ranges imply a northward transport of 14.0° + 3.7° or − 3.9° in latitude, or about 1500 ± 400 km since 16 ± 1 m.y. This implies a net transport rate of about 9.4 cm/yr. Since this rate is about two times faster than the fastest rate calculated by *Atwater and Molnar* [1973] for Pacific-North American plate motion (which is northwest not due north) and implies over 1000 km more northward displacement since 16 m.y. ago, these data are clearly contradictory. The maximum amount of latitude change since 16 m.y. ago should be about 4° ± 3° using the most generous error limits suggested by *Atwater and Molnar* [1973]. This amount of transport has been supported by paleomagnetic data from Deep Sea Drilling Project cores [*Weinreich et al.*, 1983] which indicate a northward translation of the Pacific plate of about 5° since 16 m.y. This leaves a discrepancy of 10° of northward latitudinal displacement for the northern Channel Islands.

An offset dipole or significant nondipole field components could account for an inclination shallowing of about 4° to 5° [*Wilson*, 1971; *Coupland and Van der Voo*, 1980; *Epp et al.*, 1983]. An undetected initial dip for the lava flows could also account for a few degrees of inclination shallowing. The predominant dip of the units we studied is to the north [*Kamerling and Luyendyk*, 1979*a*]. An initial westward dip (northward after rotation) would tend to decrease the apparent inclination if units were corrected to horizontal [*Kamerling*, 1980]; this would also decrease the declination anomaly. The shallow inclinations found for the tuff beds on Santa Rosa Island or for the Eocene sandstones from San Miguel and Santa Cruz Islands presumably would not be affected by a problem with undetected initial dip. However, the very high precision parameter for the tuffs (Table 4*a*) suggests they were deposited hot inferring that they too could have been deposited with an initial dip. The Eocene sandstone results are very poor in any case. It is interesting that the results from San Miguel, Santa Cruz, and Anacapa Island [*Kamerling and Luyendyk*, 1979*a*], which are predominantly from volcanic flows, are about 15° shallower in inclination than the results from Santa Rosa Island and the Santa Monica Mountains [*Kamerling and Luyendyk*, 1979*a*] which are from intrusive units. We also calculated *Kono*'s [1980] statistics for all the flows (29) and all intrusives (23) we studied; the flows have a mean paleomagnetic inclination of 32.4° (alpha 95 = 7.07°, kappa = 14.8), while the intrusives have an inclination of 40.6° (alpha 95 = 6.8°, kappa = 20.0). The directions from the intrusive units presumably do not have a problem with undetected initial dip since attitudes of sedimentary strata were used for the structural corrections.

Shallow inclinations reported by *Marshall et al.* [1979] and *Pischke et al.* [1979] from Miocene volcanic rocks in southern and Baja California suggest about 10° of northward motion. *Calderone and Butler* [1984] found significant flattening of inclination for early and middle Miocene age lavas from southwest Arizona which could imply similar motion. *Luyendyk et al.* [this issue] point out that many Neogene paleomagnetic results from southern California and Arizona show large flattening of inclination; even those from east of the San Andreas fault or on cratonic North America according to terrane accretion models. They conclude that the inclination data lack the resolution to distinguish any relative motion between the offshore islands and North America during the Neogene.

TECTONIC MODELS OF BORDERLAND FORMATION

The tectonic models which have been proposed to account for the formation and present distribution of rocks in the southern California borderland can be conveniently divided according to the dominant tectonic process which is called upon. The processes include rifting [*Yeats*, 1968*a*, *b*; *Yeats et al.*, 1974], strike-slip [*Howell et al.*, 1974; *Howell*, 1975, 1976*b*; *Blake et al.*, 1978, *Howell and Vedder*, 1981], and rotation [*Jones et al.*, 1976; *Crouch*, 1979; *Kamerling and Luyendyk*, 1979*a*, 1981; *Kamerling et al.*, 1980; *Luyendyk et al.*, 1980; *Hall*, 1981*b*].

Since paleomagnetic data from onshore southern California and Baja California [*Luyendyk et al.*, this issue; *Marshall et al.*, 1979] display inclinations about as shallow as the present study, relative displacement between the two areas is constrained mainly by the error limits involved, which allow several hundreds of kilometers. No rigorous constraints can be put on the strike-slip models with the present paleomagnetic

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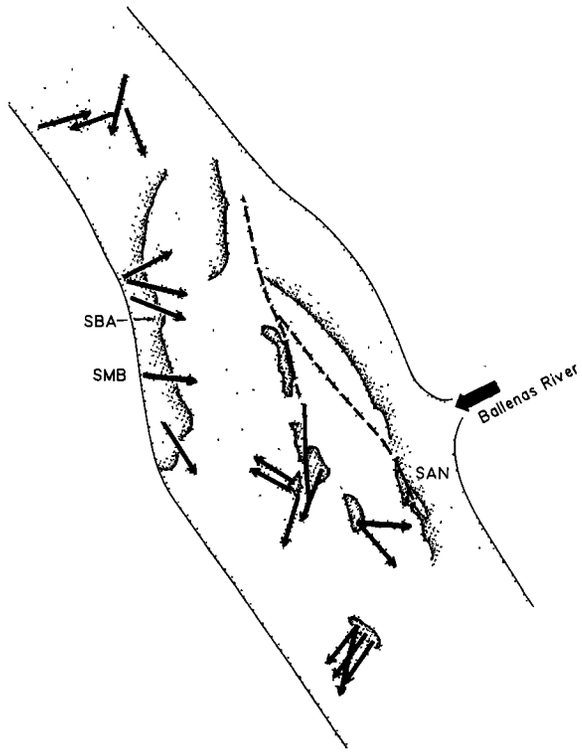


Fig. 13*a*. An Eocene basin is restored to alignment with other Eocene rocks in southern California yielding a continuous north-northwest/south-southeast trending basin (stippled). Heavy arrows are paleocurrent directions [*Howell and McLean*, 1976; *Stuart*, 1979]. The occurrences of Poway type conglomerates are brought into juxtaposition with the Ballenas River (large arrow) and its deposits near San Diego. During late Oligocene time the East Pacific Rise encountered the continental margin approximately in the area of the borderland. This resulted in uplift and extensional tectonics in the area of the Transverse Ranges. SBA, Santa Barbara; SMB, Santa Maria Basin; SAN, San Diego.

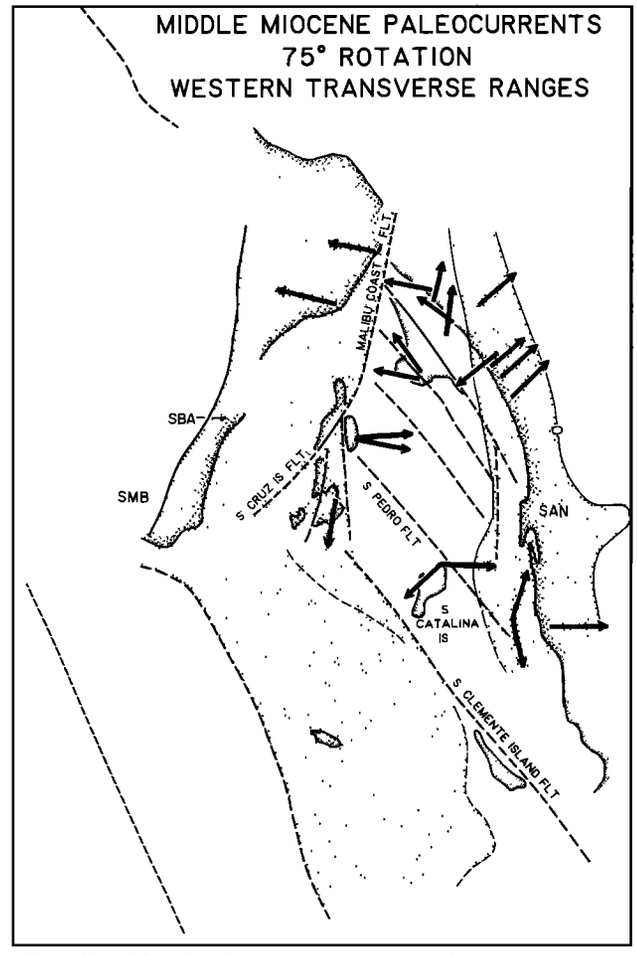


Fig. 13*b*. After the rise crest migrated away from this general location around 20 m.y. ago, northwest oriented right lateral shear between the Pacific and North American plates began to dominate the area. The previously existing trend of the extensional features was caught across this shear zone and began to rotate clockwise. The pluton on Santa Catalina Island acted as a resistant ball bearing in a matrix of more easily deformed Catalina Schist. This resulted in rotation of Santa Catalina Island [*Kamerling and Luyendyk*, 1981] between the San Clemente Island fault and the San Pedro fault as the fault bounded blocks to the west were translated northward. Heavy arrows are paleocurrent directions [*Howell and McLean*, 1976; *Stuart*, 1979].

Fig. 13. Tectonic model for the evolution of the southern California continental borderland since Eocene time. Base map modified from *Howell* [1974] and *Blake et al.* [1978].

data; neither can a purely east-west rifting model be constrained by the same data. Therefore, discussion will be focused on those models calling upon rotation.

In his reconstruction of the borderland, which is based on similar Eocene Poway type conglomerates on the northern Channel Islands and the San Diego area, *Cole* [1970] backrotated Santa Cruz Island through a large angle (115° CCW or 245° CW) in order to more favorably position the southwestward paleocurrent directions found there [*Merschat*, 1971]. Using structural lineations in the basement rocks of Santa Cruz and Santa Catalina Island, paleocurrent directions, and his model for the East Santa Cruz Basin fault system, *Howell* [1976*a*] proposed a 32° counterclockwise rotation of the southern part of Santa Cruz Island relative to the northern part. *Hall* [1981*b*] proposed a model involving counterclockwise rotation of the western Transverse Ranges and concurrent opening of the Santa Maria basin as a pull-apart structure. These models involving counterclockwise rotation of the western Transverse Ranges are difficult to reconcile with the paleomagnetic evidence. *Hornafius* [this issue] shows a consistent decrease in clockwise declination with decreasing age for sites in the middle Miocene Monterey Formation along the northern Santa Barabara Channel coast. This indicates that the rotation was clockwise. Although 283° of counterclockwise rotation could account for the mean declination anomaly, this amount of counterclockwise rotation is much larger than the amounts called for by *Hall* [1981*b*] or *Howell* [1976*a*] in their models.

The results obtained from the present study and the data of *Kamerling and Luyendyk* [1979*a*] and *Hornafius* [this issue] in

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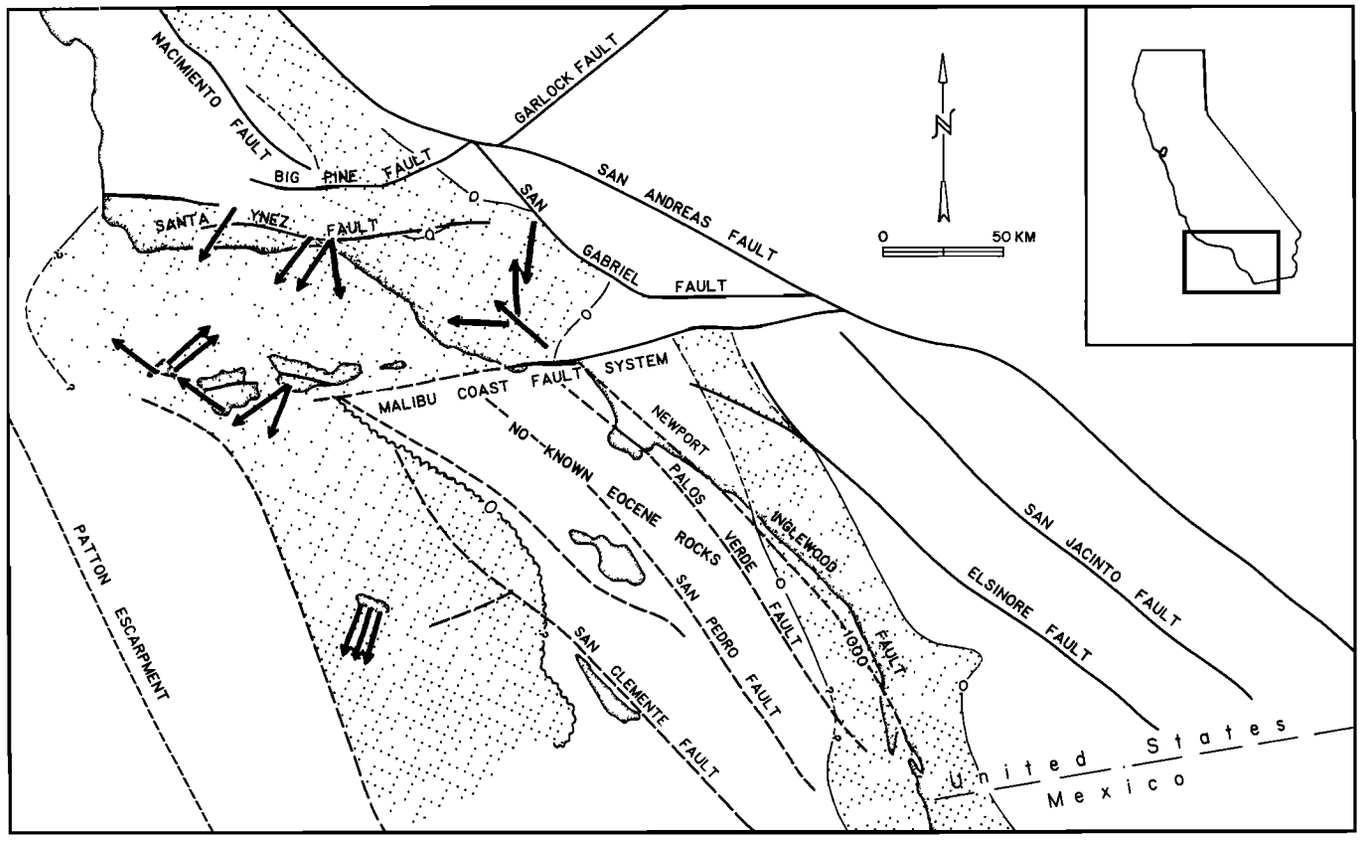


Fig. 13c. Clockwise rotation ended or slowed dramatically, when the San Gabriel/San Andreas system was initiated thereby concentrating the shear along a more narrow zone to the east. The Transverse Ranges were then compressed as the area west of the San Andreas fault moved northwestward with the Pacific plate.

addition to the geologic evidence [*Jones and Irwin*, 1975; *Crouch*, 1979; *Nilsen and Abbott*, 1981; *Sundberg and Cooper*, 1982; *Dickinson*, 1983] strongly suggest that the southern part of the western Transverse Ranges has rotated as a more or less coherent block at least 80° clockwise and possibly more, since early or middle Miocene time. Thus the data are in substantial agreement with the models calling for clockwise rotation of the western Transverse Ranges. If the effects of local counterclockwise rotation of fault blocks and rotation during the formation of the volcanics are real, then the paleomagnetically determined amount of rotation may be more of a minimum estimate of the total rotation than the actual amount. Rotation of about 100° brings Poway type rocks on the islands in juxtaposition with the Eocene Poway Conglomerate in San Diego area, if the Transverse Ranges are rotated about an eastern pivot (Figure 13*a*). The correlation of these rocks [*Cole*, 1970; *Howell et al.*, 1974; *Abbott and Smith*, 1978; *Howell and Link*, 1979; *Abbott et al.*, 1983] seems to demand the juxtaposition. This reconstruction would put San Nicolas Island south of the San Diego area, a location which fits the provenance data of *Abbott et al.* [1983]. The correlation of Cretaceous submarine fan facies rocks on San Miguel Island and in the San Diego and northern Baja California area [*Bartling*, 1981; *Bartling and Abbott*, 1983; *Dickinson*, 1983; *Nilsen and Abbott*, 1981; *Sundberg and Cooper*, 1982] also fit such a reconstruction. Northwest to northeast directed paleocurrent directions on San Miguel Island and Santa Rosa Island [*Parseley*, 1972; *Weaver*, 1969] would be reoriented toward the west, compatible with a submarine fan model for these rocks. However, the southwest directed paleocurrent directions [*Merschat*, 1971] in the correlative Eocene rocks from the southern part of Santa Cruz Island would be reoriented toward the southeast. If the southern part of Santa Cruz Island was repositioned south of San Miguel Island then this paleocurrent direction is more satisfactory. Considering the anomalous paleomagnetic results and the stratigraphic differences between the northern and southern parts of the island it seems reasonable to assume that the southern part of Santa Cruz Island has moved independently from the rest of the islands.

Except for the local rotations discussed above and the large apparent rotations the southern part of Santa Cruz Island, the consistency of the data suggest that the southern part of the western Transverse Ranges rotated by about the same amount probably as a more or less coherent block (Figures 13*a*–13*c*). This coherent block was probably held intact by a previously developed fault pattern and increased coupling between the crust and a viscous high seismic velocity upper mantle ridge beneath the Transverse Ranges [*Hadley and Kanamori*, 1977]. This earlier fault pattern and possibly the mantle ridge may have resulted from propagation of the East Pacific Rise into the continental margin between 29 and 21 m.y. The plate reconstructions of *Atwater and Molnar* [1973] suggest that the rise remained in about the same position relative to North America during this interval. Although there is some uncertainty in their reconstruction, this position appears to have

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been close to the position of the Transverse Ranges restored for the opening of the Gulf of California. *Lee* [1973] also suggested that the East Pacific Rise intersected the continental margin near the Transverse Ranges based on heat flow data and theoretical models. We suggest that after the rise crest had migrated south this elongate block was caught across a widening shear zone between the North American and Pacific plates and rotated clockwise in the manner proposed by *Luyendyk et al.* [1980]. As the Transverse Range block rotated, rifting occurred in the inner borderland and Catalina Schist basement welled up and was shed outward from the rifted area [*Kamerling and Luyendyk*, 1979*b*; *Dickinson*, 1983]. Rotation probably continued until the Transverse Range block was oriented east-west (Figure 13*c*) and the shear between the Pacific and North American plates was concentrated to the east along the San Andreas fault. This occurred around 5 m.y. [*Hornafius et al.*, 1982], although rotation may have only slowed dramatically at this time and still be continuing [*Hornafius*, this issue]. The orientation of fold axes and faults at the far western end of the Channel Islands platform curve to the northwest suggesting that this area may have been, or is still being, rotated more than the eastern part of the island platform. However, the paleomagnetic data from the islands cannot substantiate this idea.

SUMMARY AND CONCLUSIONS

Late Oligocene through middle Miocene volcanic rocks from the northern Channel Islands appear to have faithfully recorded the geomagnetic field although structural complexities have increased the dispersion of the data in some areas. The southern Transverse Ranges area probably rotated as a coherent unit within the Pacific-North American dextral shear zone. However, a local left-shear couple [*Luyendyk et al.*, 1980] may have caused small (tenths, to a few km^2) fault blocks to rotate counterclockwise within the larger clockwise rotating unit. When compared to the 20 Ma reference pole for North America the data suggest a net clockwise tectonic rotation of the southern part of the western Transverse Ranges of $76.1° \pm 7.0°$ and a flattening of inclination of $18.2° \pm 5.7°$. Due to the effects of counterclockwise local block rotations this amount of rotation may be a minimum for the region as a whole. The inclination data suggest a northward movement of $14.0° + 3.7°$ or $- 3.9°$ for this area which is large compared to most plate motion models for the period. Nondipole field behavior, an offset dipole, and possible undetected initial dip of lava flows, could explain part of the observed flattening. Because the paleomagnetic data from south California and Baja California and southwest Arizona also indicate about 10° of northward motion [*Luyendyk et al.*, this issue; *Marshall et al.*, 1979; *Calderone and Butler*, 1984], relative motion between the Transverse Ranges and North America could range from 8° to zero within the confidence limits.

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REFERENCES

Abbott, P. L., and T. E. Smith, Trace-element comparison of clasts in Eocene conglomerates, southwestern California and northwestern Mexico, *J. Geol.*, *86*, 753–762, 1978.

Abbott, P. L., R. P. Kies, and W. R. Bachman, A tectonic slice of Eocene strata, northern part of California continental borderland, in *Cenozoic Marine Sedimentation, Pacific Margin, U.S.A.*, edited by D. K. Larue and R. J. Steel, pp. 151–168, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1983.

Anderson, R. E., L. E. Redwine, and P. E. McGovney, Geology of northern Santa Rosa Island, *Am. Assoc. Pet. Geol. Bull.*, *33*, 2062, 1949.

Atwater, T., and P. Molnar, Relative motion of the Pacific and North American plates deduced from seafloor spreading in The Atlantic, Indian, and South Pacific oceans, *Stanford Univ. Publ. Geol. Sci.*, *13*, 136–148, 1973.

Avila, F. A., and D. W. Weaver, Mid-Tertiary stratigraphy, Santa Rosa Island, in *Geology of the Northern Channel Islands*, edited by D. W. Weaver, pp. 48–67, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1969.

Baird, A. K., D. M. Morton, A. O. Woodford, and K. W. Baird, Transverse Ranges province: A unique structural-petrochemical belt across the San Andreas fault system, *Geol. Soc. Am. Bull.*, *85*, 163–174, 1974.

Bartling, W. A., Sedimentology of Upper Cretaceous strata, San Miguel Island, California, and its relation to the tectonics of the southern California borderland, M.S. thesis, San Diego State Univ., San Diego, Calif., 1981.

Bartling, W. A., and P. L. Abbott, Upper Cretaceous sedimentation and tectonics with reference to the Eocene, San Miguel Island, and San Diego area, California, in *Cenozoic Marine Sedimentation Pacific Margin, U.S.A.*, edited by D. K. Larue and R. J. Steel, pp. 133–150, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1983.

Beck, M. E., Jr., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res.*, *85*(B12), 7115–7131, 1980.

Bereskin, S. R., and L. N. Edwards, Mid-Tertiary stratigraphy, southwestern Santa Cruz Island, in *Geology of the Northern Channel Islands*, edited by D. W. Weaver, pp. 68–79, Pacific Section, Society of Economic Paleontologists Mineralogists, Los Angeles, Calif., 1969.

Blake, M. C., Jr., R. H. Campbell, T. W. Dibblee, Jr., D. G. Howell, T. H. Nilsen, W. R. Normark, J. C. Vedder, and E. A. Silver, Neogene basin formation in relation to plate-tectonic evolution of San Andreas fault system, California, *Am. Assoc. Pet. Geol. Bull.*, *62*, 344–372, 1978.

Calderone, G., and R. F. Butler, Paleomagnetism of Miocene volcanic rocks from southwestern Arizona: Tectonic implications, *Geology*, *12*, 627–630, 1984.

Champion, D. E., D. G. Howell, and M. C. Marshall, Paleomagnetic evidence for 3800 km of northwestward translation of San Miguel Island, southern California borderland (abstract), *Eos Trans. AGU*, *62*, 855, 1981.

Cloos, E., Experimental analysis of fracture patterns, *Geol. Soc. Am. Bull.*, *66*, 241–256, 1955.

Cole, M. R., Paleocurrent and basin analysis on San Nicolas Island, California, M. S. thesis, Ohio Univ., Athens, 1970.

Cole, M. R., Eocene paleocurrents and sedimentation, San Nicolas Island, California, *Am. Assoc. Pet. Geol. Bull.*, *61*, 237–247, 1977.

Coupland, D. H., and R. Van der Voo, Long-term nondipole components in the geomagnetic field during the last 130 m.y., *J. Geophys. Res.*, *85*, 3529–3548, 1980.

Crouch, J. K., Neogene tectonic evolution of the California continental borderland and western Transverse Ranges, *Geol. Soc. Am. Bull.*, *90*, 338–345, 1979.

Crowe, B. M., H. McLean, D. G. Howell, and R. E. Higgins, Petrography and major-element chemistry of the Santa Cruz Island Volcanics, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 196–215, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Demarest, H. H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. Geophys. Res.*, *88*, 4321–4328, 1983.

Dibblee, T. W., Jr., Geology of southwestern Santa Barbara County, California, *Bull. Calif. Div. Mines Geol.*, *150*, 95 pp., 1950.

Dibblee, T. W., Jr., Strike-slip tectonics of the San Andreas fault and

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its role in Cenozoic basin evolvement, in *Late Mesozoic and Cenozoic Sedimentation and Tectonics in California*, pp. 26–38, San Joaquin Geological Society, Bakersfield, Calif., 1977.

Dickinson, W. R., Cretaceous sinistral strike-slip along Nacimiento fault in coastal California, *Am. Assoc. Pet. Geol. Bull., 67*, 624–645, 1983.

Doerner, D. P., Lower Tertiary biostratigraphy of southwestern Santa Cruz Island, in *Geology of the Northern Channel Islands*, edited by D. W. Weaver, pp. 17–29, Pacific Section, Society of Economic Paleontologists and Mineralogists, 1969.

Epp, D., W. W. Sager, F. Theyer, and S. R. Hammond, Hotspot-spin axis motion or magnetic far-sided effect?, *Nature, 303*, 318–320, 1983.

Erickson, J. W., Petrology of some middle and late Eocene sandstones from the southern California borderland, in *Paleogene Symposium and Selected Technical Papers*, edited by D. W. Weaver et al., pp. 191–209, Amer. Association of Petroleum Geologists, Tulsa, Okla., 1975.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217*, 295–305, 1953.

Fisher, R. V., and D. W. Charlton, Mid-Miocene Blanca Formation, Santa Cruz Island, California, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 228–240, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Hadley, D., and H. Kanamori, Seismic structure of the Transverse Ranges, California, *Geol. Soc. Am. Bull, 88*, 1469–1478, 1977.

Hall, C. A., Jr., Origin and development of the Lompoc-Santa Maria pull-apart Basin and its relation to the San Simeon-Hosgri strike-slip fault, western California, *Spec. Rep. Calif. Div. Mines Geol., 137*, 25–31, 1978.

Hall, C. A., Jr., San Luis Obispo transform fault and middle Miocene rotation of the western Transverse Ranges, California, *J. Geophys. Res., 86*(B2), 1015–1031, 1981*a*.

Hall, C. A., Jr., Evolution of the western Transverse Ranges microplate: Late Cenozoic faulting and basinal development, in *The Geotectonic Development of California*, edited by W. G. Ernst, pp. 559–582, Prentice-Hall, Englewood Cliffs, N.J., 1981*b*.

Hill, D. J., Geology of the Jurassic basement rocks, Santa Cruz Island, California, and correlation with other Mesozoic basement terranes in California, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 16–46, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Hornafius, J. S., Neogene tectonic rotation of the Santa Ynez Range, western Transverse Ranges, California, suggested by paleomagnetic investigation of the Monterey Formation, *J. Geophys. Res.*, this issue.

Hornafius, J. S., B. P. Luyendyk, D. W. Weaver, W. W. Wornardt, and M. Fuller, Paleomagnetic results from the Monterey Formation in the western Transverse Ranges (abstract), *Eos Trans. AGU, 62*, 855, 1981.

Hornafius, S., M. Kamerling, R. Terres, and B. Luyendyk, Differential tectonic rotation within the western Transverse Ranges, *Geol. Soc. Am. Abstr. Programs, 14*(4), 173–174, 1982.

Howell, D. G., Middle Eocene paleography of southern California, Ph.D. thesis, Univ. of Calif., Santa Barbara, 1974.

Howell, D. G., Hypothesis suggesting 700 km of right-slip in California along northwest-oriented faults, *Geology, 3*(2), 81–83, 1975.

Howell, D. G., Late-Miocene counterclockwise rotation of the south half of Santa Cruz Island, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 449–454, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*a*.

Howell, D. G., A model to accommodate 1000 kilometers of right-slip, Neogene, displacement in the southern California area, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 530–540, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*b*.

Howell, D. G., and M. H. Link, Eocene conglomerate sedimentology and basin analysis, San Diego and the southern California borderland, *J. Sediment. Petrol., 49*, 517–540, 1979.

Howell, D. G., and H. McLean, Middle Miocene paleogeography, Santa Cruz and Santa Rosa Islands, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 266–293, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Howell, D. G., and J. G. Vedder, Structural implications of stratigraphic discontinuities across the southern California borderland, in *The Geotectonic Development of California*, edited by W. G. Ernst, pp. 535–558, Prentice-Hall, Englewood Cliffs, N.J., 1981.

Howell, D. G., C. J. Stuart, J. P. Platt, and D. J. Hill, Possible strike slip faulting in the southern California borderland, *Geology, 2*(2), 93–98, 1974.

Ingle, J. C., Jr., Cenozoic paleobathymetry and depositional history of selected sequences within the southern California continental borderland, *Spec. Publ. Cushman Found. Foraminiferal Res. 19*, 163–195, 1980.

Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terrains, *Can. J. Earth Sci., 16*, 669–694, 1979.

Jones, D. L., and W. P. Irwin, Rotated Jurassic rocks in the Transverse Ranges, California, *Geol. Soc. Am. Abstr. Programs, 7*, 330, 1975.

Jones, D. L., M. C. Blake, and C. Rangin, The four Jurassic belts of northern California and their significance to the geology of the southern California borderland, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 342–362, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Junger, A., Offshore structure between Santa Cruz and Santa Rosa Islands, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 428–442, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*a*.

Junger, A., Tectonics of the southern California borderland, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 486–498, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*b*.

Junger, A., Maps and seismic profiles showing geologic structure of the northern Channel Islands Platform, continental borderland, *U.S. Geol. Surv. Misc. Field Stud., Map MF-991*, 1979.

Kamerling, M. J., Paleomagnetism of middle Miocene volcanics and tectonic rotation of the Santa Monica Mountains region, western Transverse Ranges, California, M.A. thesis, Univ. of Calif., Santa Barbara, 1980.

Kamerling, M. J., and B. P. Luyendyk, Tectonic rotation of the Santa Monica Mountains in southern California (abstract), *Eos Trans. AGU, 58*, 1126, 1977.

Kamerling, M. J., and B. P. Luyendyk, Tectonic rotations of the Santa Monica Mountains region, western Transverse Ranges, California, suggested by paleomagnetic vectors, *Geol. Soc. Am. Bull., 90*, 331 337, 1979*a*.

Kamerling, M. J., and B. P. Luyendyk, A model for Neogene tectonics of the inner southern California borderland constrained by paleomagnetic data, *Geol. Soc. Am. Abstr. Programs, 11*, 453, 1979*b*.

Kamerling, M., and B. Luyendyk, Paleomagnetism and tectonics of the islands of the southern California continental borderland (abstract), *Eos Trans. AGU, 62*, 855, 1981.

Kamerling, M. J., B. P. Luyendyk, and M. Marshall, Paleomagnetism and tectonic rotation of parts of the Transverse Ranges, California (abstract), *Eos Trans. AGU, 59*, 1058, 1978.

Kamerling, M. J., B. P. Luyendyk, T. S. Powell, and R. Terres, Tectonic rotation of the northern Channel Islands of the Southern California borderland (abstract), *Eos Trans. AGU, 61*, 1125, 1980.

Kies, R. P., and P. L. Abbott, Rhyolite clast populations and tectonics in the California continental borderland, *J. Sediment. Petrol. 53*, 461–475, 1983.

Kono, M., Statistics of paleomagnetic inclination data, *J. Geophys. Res., 85*(B7), 3878–3882, 1980.

Lee, T. C., Heat flow and other geophysical studies in the southern California borderland, Ph.D. dissertation, Univ. of South. Calif., Los Angeles, 1973.

Luyendyk, B. P., M. J. Kamerling, and R. Terres, Geometric model for Neogene crustal rotations in southern California, *Geol. Soc. Am. Bull., 91*, 211–217, 1980.

Luyendyk, B. P., E. C. Hajic, R. Crippen, and D. S. Simonett, Side scan sonar and high resolution reflection maps of the Santa Barbara Channel seafloor, contract T-CSG-CP-00, 17 pp., Sea Grant Coll. Program, Univ. of Calif., La Jolla, 1983.

Luyendyk, B. P., M. J. Kamerling, R. R. Terres, and J. S. Hornafius, Simple shear of southern California during Neogene time suggested by paleomagnetic declinations, *J. Geophys. Res.*, this issue.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, *J. Geophys. Res. 85*, 3659–3669, 1980.

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12485 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

MacKinnon, T. C., Cretaceous sedimentation, southern San Rafael Mountains, Santa Barbara County, California, M.A. thesis, Univ. of Calif., Santa Barbara, 1976.

Marshall, M., J. Callian, G. Pischke, N. Mace, and G. Gastill, Paleomagnetic evidence for possible major right lateral Neogene shear in western Mexico and Arizona (abstract), *Eos Trans. AGU*, *60*, 817, 1979.

McLean, H., B. M. Crowe, and D. G. Howell, Source of Blanca Formation volcaniclastic rocks and strike-slip faulting on Santa Cruz Island, California, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 294–308, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*a*.

McLean, H., D. G. Howell, and J. G. Vedder, Miocene strata on Santa Cruz and Santa Rosa Islands—A reflection of tectonic events in the southern California borderland, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 241–253, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976*b*.

Merschat, W. R., Lower Tertiary paleocurrent trends, Santa Cruz Island, California, M.S. thesis, Ohio Univ., Athens, 1971.

Minch, J. A., K. N. Gibson, and G. L. Peterson, Clast populations in Sespe and Poway conglomerates and their possible bearing on the tectonics of the southern California borderland, Aspects of the Geologic History of the California Continental Borderland, *Misc. Publ. 24*, edited by D. G. Howell, pp. 256–265, Am. Assoc. of Pet. Geol., Tulsa, Okla., 1976.

Ness, G. E., S. Levi, and R. W. Couch, Marine magnetic anomaly time scales for the Cenozoic and late Cretaceous: A precis, critique, and synthesis, *Rev. Geophys.*, *18*, 753–770, 1980.

Nilsen, T. H., and P. L. Abbott, Paleogeography and sedimentology of Upper Cretaceous turbidites, San Diego, California, *Am. Assoc. Pet. Geol. Bull.*, *65*, 1256–1284, 1981.

Nolf, B., and P. Nolf, Santa Cruz Island volcanics, in *Geology of the Northern Channel Islands*, edited by D. W. Weaver, pp. 91–94, Pacif. Sect., Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1969.

Parsley, R. M., Late Cretaceous through Eocene paleocurrent directions, paleoenvironment and paleogeography of San Miguel Island, California, M.S. thesis, Ohio Univ., Athens, 1972.

Pischke, G. M., C. M. Marshall, and R. G. Gastil, Miocene paleomagnetic study of the tectonic history of Baja California, Mexico, *Geol. Soc. Am. Abstr. Programs*, *11*, 122, 1979.

Stuart, C. J., Middle Miocene paleogeography of coastal southern California and the California borderland—Evidence from schist-bearing sedimentary rocks, in *Cenozoic Paleogeography of the Western United States, Pacific Coast Paleogeography Symposium 3*, edited by J. M. Armentrout, M. R. Cole, and H. Terbest, Jr., pp. 29–44, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1979.

Sundberg, F. A., and J. D. Cooper, Late Cretaceous (Campanian) paleogeography of southern California, in *Late Cretaceous Depositional Environments and Paleogeography, Santa Ana Mountains, Southern California*, edited by D. J. Bottjer, I. P. Colburn, and J. D. Cooper, pp. 115–121, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1982.

Terres, R. R., M. J. Kamerling, and B. P. Luyendyk, Major clockwise rotation of the Diligencia Basin of southeastern California (abstract), *Eos Trans. AGU*, *61*, 1125, 1980.

Turner, D. L., Potassium-argon dating of Pacific Coast Miocene foraminiferal stages, Radiometric Dating and Paleontologic Zonation, edited by O. L. Bandy, *Spec. Pap. Geol. Soc. Am.*, *124*, 91–129, 1970.

Weaver, D. W., Paleogeographic implications and geologic history [of the northern Channel Islands], in *Geology of the Northern Channel Islands*, edited by D. G. Weaver, pp. 115–124, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1969.

Weaver, D. W., and D. P. Doerner, Lower Tertiary stratigraphy, San Miguel and Santa Rosa Islands, in *Geology of the Northern Channel Islands*, edited by D. W. Weaver, pp. 30–47, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1969.

Weinreich, N., F. Theyer, and Scientific Staff, Leg 85 of the DSDP/IPOD, Success and failure of HPC-magnetostratigraphy of DSDP leg 85, central equatorial Pacific (abstract), *Eos Trans. AGU*, *64* 219, 1983.

Wilcox, R. E., T. P. Harding, and D. R. Seely, Basic wrench tectonics, *Am. Assoc. Pet. Geol. Bull.*, *57*, 74–96, 1973.

Wilson, R. L., Dipole offset—The time average paleomagnetic field over the past 25 million years, *Geophys. J. R. Astron. Soc.*, *22*, 491–504, 1971.

Yeats, R. S., Rifting and rafting in the southern California Borderland, in *Stanford Univ. Publ. Geol. Sci.*, 307–322, 1968*a*.

Yeats, R. S., Southern California structure, sea-floor spreading and history of the Pacific basin, *Geol. Soc. Am. Bull.*, *79*, 1693–1702, 1968*b*.

Yeats, R. S., M. R. Cole, W. R. Merschat, and R. M. Parsley, Poway fan and submarine cone and rifting of the inner southern California borderland, *Geol. Soc. Am. Bull.*, *85*, 293–302, 1974.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. W. Collison, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, New York, 1967.

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